Tara 11-11-26 Conversation 1: Tape spring L001 [0/0/0] - axial, bending and coiling Hashin checks, mass, PDF report

Claude Code session, 11-11-2026, 11:23 - 16:38 PST. User messages and Claude replies in full; tool calls and outputs shortened; Claude's internal reasoning omitted.

User · 11:23

I have a tape spring boom with the following properties: material L001 in the material property catalog, layup [0/0/0], radius of curvature 20 mm, subtended angle theta = 180 degrees, length 1 meter, cantilever boom clamped at 1 end. Use the Python tools in the .claude/skills folder to write and run OpenSeesPy code to evaluate the following load cases: • Axial extension load of 10 N • Transverse tip load of 1 N in opposite-sense bending (report the tip deflection) • Coiling around a hub with 4 cm radius (the code for this should be in the failure-check skill) For each case, report how close the boom is to failure using the Hashin criteria in the failure-check skill. Also calculate the mass of this boom using the code to calculate mass of a boom in the objective-constraints skill. Provide a status update every 15 minutes. Only use 16 CPU cores, and do not modify any files do not modify any files on this server outside the F drive / Tara / BenazzoVenkatadri_ThinShellSims folder. Put all code you create during this optimization in Code for Project / 11/11/26 (Tara).

update_plan task list

```
[in_progress] Read skill docs and catalogs
[pending] Write OpenSeesPy model scripts
[pending] Run axial, bending and coiling cases
[pending] Hashin failure checks per case
[pending] Compute boom mass
[pending] Report results
```

```
Plan shown (6 steps).
```

Bash List skills, project code folders, and L001 catalog entry

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; ls .claude/skills/*; ls "Code for Project"; ls .claude/orchestrator; cat
.claude/orchestrator/ACTIVE_STUDY 2>/dev/null; grep -n "L001" .claude/memory.material_property_catalog/*.txt | head
```

```
.claude/skills/README.md

.claude/skills/abaqus-export:
SKILL.md
case_to_abaqus.py

.claude/skills/design-boom:
SKILL.md

.claude/skills/failure-check:
[... shortened ...]
```

Bash Inspect target folder and catalog header

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; ls -la "Code for Project/11-11-26 (Tara)"; sed -n 1,16p
.claude/memory.material_property_catalog/UD_lamina_catalog_SI.txt; sed -n 75,90p
.claude/memory.material_property_catalog/UD_lamina_catalog_SI.txt; sed -n 100,117p
.claude/memory.material_property_catalog/UD_lamina_catalog_SI.txt | cut -c1-200
```

```
total 4
drwxr-xr-x 1 tvenkata 1049089 0 Nov 11 11:22 .
drwxr-xr-x 1 tvenkata 1049089 0 Nov 11 11:05 ..
# UD lamina property catalog (SI units: Pa, m, kg/m3). GENERATED from
# UD_lamina_catalog_readable.txt by lamina_catalog_to_SI.py; edit the readable file, then rerun.
# Notes, estimate markers (*) and sources below are those of the readable file.
# Maintained by the CompositeLaminaLibrarian agent. SI twin: UD_lamina_catalog_SI.txt
# (generated from this file by lamina_catalog_to_SI.py; do not edit the SI file by hand).
#
# Columns: ID | Fiber | Polymer system | E1 | E2 | G12 | nu12 | nu23 | G23 | Xc | Xt | Yc | Yt | S12 | t_ply | rho | Source
[... shortened ...]
```

Claude · 11:24

Now the skill docs.

Bash Read skills README, failure-check and objective-constraints docs

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills"; cat README.md; cat failure-check/SKILL.md; cat objective-constraints/SKILL.md
```

```
# Thin-shell simulation skills (v2.0)

Five scripts replace the seven v1.0 agents (`.claude/agents_v1.0/`) and the
repeated model code in `Code for Project/`. Each folder has a `SKILL.md`
(instructions) and one Python script. The orchestrator agent
(`.claude/agents_v2.0/ThinShellOrchestrator.md`) runs them in order;
`CompositeLaminaLibrarian` (v2.0) still maintains the material catalog; the
opensees-to-abaqus skill (11/4/26) replaces the `OpenSeesPyToAbaqusPythonGenerator`
agent and turns an assembled OpenSeesPy script into an Abaqus/CAE script.

[... shortened ...]
```

Bash Search failure_check.py for hub/coiling options

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills"; grep -n -i "hub\|coil\|argparse\|add_argument" failure-check/failure_check.py | head -80; wc -l */*.py
```

```
50:Coiling check (only if the simulation gives F < 1): flattening a flange of
57:coiling F is above the simulation F it is reported as the governing F.
67:import argparse
293:# --- coiling (flattening) check ----------------------------------------------------
310:def coiling_check(lam, r, overrides=None, criterion='hashin', S23=None):
363:def coiling_report(lam, r, src, out, criterion):
370:        f'Coiling check: flange [{lam.describe()}] flattened from r = {r:g} mm ({src}): '
380:    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
381:    ap.add_argument('results', help='.npz from shell-bc-load')
382:    ap.add_argument('--criterion', choices=('hashin', 'tsai-hill'), default='hashin')
[... shortened ...]
```

Bash Read failure_check.py header and coiling section

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills"; sed -n 1,66p failure-check/failure_check.py; sed -n 290,464p failure-check/failure_check.py
```

```
"""
Ply stresses in every element and the 2D Hashin failure criterion (default;
Tsai-Hill as an option) of a solved shell model (results .npz from the
shell-bc-load skill, laminate and strengths from the laminate-properties skill).

Ply stresses (classical lamination theory kinematics on the FE strains):
for each facet and Gauss point the generalized strains (eps0, k) in the
facet frame give eps(z) = eps0 + z k at the bottom, middle and top of every
ply; rotated to the ply axes (angle theta from the facet x axis) and
multiplied by the ply's reduced stiffness Q:
[... shortened ...]
```

Bash Find project's opposite-sense bending convention

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; grep -rn -i "opposite.sense\|opposite-sense\|equal.sense" .claude/orchestrator/DOMAIN_KNOWLEDGE.md .claude/skills/*/SKILL.md .claude/skills/*/*.py | head -30
```

```
.claude/orchestrator/DOMAIN_KNOWLEDGE.md:52:| Typical failure / instability | snap-through (opposite sense), torsional buckling (equal sense), flexural-torsional column buckling | flange-tip local buckling and lateral-torsional buckling | local buckling of the flattest arc; tab delamination; strongly imperfection-sensitive | local lobe buckling |
.claude/orchestrator/DOMAIN_KNOWLEDGE.md:72:  - *Opposite-sense bending* (longitudinal curvature opposite in sign to the
.claude/orchestrator/DOMAIN_KNOWLEDGE.md:79:  - *Equal-sense bending* (the edges in compression): lower peak; a
.claude/orchestrator/DOMAIN_KNOWLEDGE.md:117:  combinations (equal/opposite-sense coiling, either face out).
.claude/skills/shell-buckling/SKILL.md:40:  column, and a spurious early crossing in opposite-sense tape bending.
.claude/skills/shell-buckling/SKILL.md:58:   control for limit points (opposite-sense bending, local buckling),
```

Bash Read domain knowledge on bending senses and coiling

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; sed -n 60,125p .claude/orchestrator/DOMAIN_KNOWLEDGE.md
```

```
tape-spring arc, at about e = 2R (sin a - a cos a)/(a - sin a cos a) from
  the centre, a = alpha/2). A transverse tip load not through it twists the
  boom: in FE the "bending" tip displacement then includes twist. The
  project loads through a rigid end at the section reference point, so a
  stiffness noticeably below beam theory can be twist. Look at the PNG.
- TRAC: the web gives depth (EI about one axis) and the flanges flatten
  against it. Its weakness is lateral-torsional buckling.

## 3. Nonlinear behaviour and buckling of thin curved shells

[... shortened ...]
```

Bash Find hub coiling code and bending example

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude"; grep -rln "F_hub\|hub_check\|def .*hub" --include=*.py . | head; ls
skills/shell-bc-load/examples skills/shell-buckling/examples; cat skills/shell-
buckling/examples/tape_spring_opposite_bending.json
```

```
./orchestrator/tools/design_screen.py
./orchestrator/tools/study.py
skills/shell-bc-load/examples:
plate_tip_load_laminate.json
tape_spring_L024_tip_load.json

skills/shell-buckling/examples:
tape_spring_compression.json
tape_spring_opposite_bending.json
{
[... shortened ...]
```

Claude · 11:24

Reading the skill docs and the example tape-spring case; next I'll write the three load-case files in `Code for Project/11-11-26 (Tara)`.

Bash Read tip-load example, hub code, shell-bc-load docs

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude"; cat skills/shell-bc-load/examples/tape_spring_L024_tip_load.json; grep
-n "F_hub\|def .*hub" orchestrator/tools/design_screen.py | head; sed -n 1,200p skills/shell-bc-load/SKILL.md
```

```
{
 "name": "tape_spring_L024_tip_load",
 "laminate": {"catalog": "L024", "layup": "[0/90]s"},
 "geometry": {"L": 500, "section": {"type": "arc", "R": 50, "b": 100}, "size": 5},
 "element": {"type": "ASDShellQ4", "corotational": false},
 "supports": [{"edge": "x0", "dofs": [1, 1, 1, 1, 1, 1]}, {"rigid_end": "xL"}],
 "loads": [{"node": {"rigid": "xL"}, "force": [0, 0, -10]}],
 "analysis": {"type": "linear"},
 "report_node": {"rigid": "xL"}
}
[... shortened ...]
```

Bash Read hub coiling code, assemble-run docs, arc convention

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude"; sed -n 220,240p orchestrator/tools/design_screen.py; cat skills/shell-
assemble-run/SKILL.md | head -80; grep -n "arc" skills/shell-mesh/SKILL.md | head -20
```

```
"""max Hashin 2D over plies / through-thickness points for curvature arrays kx, ky (same shape)."""
        kx, ky = np.asarray(kx, float)[..., None, None, None], np.asarray(ky, float)[..., None, None, None]
        s = kx * self.Sx + ky * self.Sy
        a, b = fc.hashin_modes(s[..., 0], s[..., 1], s[..., 2], *self.st)
        return (a + b).max(axis=0).reshape(np.shape(kx)[:-3] + (-1,)).max(axis=-1)


def coiling_indices(L, r, R_hub):
    """F_flat (flatten only) and F_hub (flatten + hub, 4 sign combinations).
    r may be an array of radii (one laminate, many sections): arrays returned."""
[... shortened ...]
```

Bash Arc geometry convention, bending sign, env path

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude"; sed -n 70,100p skills/shell-mesh/SKILL.md; sed -n 1,80p skills/shell-buckling/SKILL.md | grep -n -i "sign\|bending" ; grep -n "sign" skills/shell-buckling/boom_buckling.py | head -20; ls "../Code for Project/11-10-26 (Tara)" | head -30; cat ~/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/memory/shells-env-path-this-machine.md
```

```
Exit code 1
piece is not that flange laminate).
It writes the nodes and the facet table, and builds the elements at run time
from `SECTIONS` of the laminate-properties piece exactly as `build_model`
(same `-local` vectors, `-noeas` rule); it provides `FACETS`, `FRAMES`, `MESH`.

## Geometry conventions
- Extrusion along x from 0 to L. Cross-section parametrized by arc length s.
- `arc(R, b)`: symmetric about y = 0, z = R (1 - cos(phi)); R > 0 = centre of
  curvature on +z. `flat(b)`: y from 0 to b. Identical nodes/tags to v1 StripMesh.
[... shortened ...]
```

Read C:\Users\tvenkata\.claude\projects\F--Tara-BenazzoVenkatadri-ThinShellSims\memory\shells-env-path-this-machine.md

```
1       ---
2       name: shells-env-path-this-machine
3       description: "On Tara's F:-drive Windows Server machine, the shells conda env is at C:\\Users\\tvenkata\\miniforge3,
not the D:\\ path the skill docs give"
4       metadata:
5         node_type: memory
6         type: project
7         originSessionId: 688ee0d2-ba68-4419-87dd-3df9c874d897
8         modified: 2026-11-07T19:24:26.519Z
9       ---
10
[... shortened ...]
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 2 of 6 is in progress.
```

Bash Write axial and bending case files; check laminate CLI

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
cat > ts_L001_axial.json <<'EOF'
{
 "name": "ts_L001_axial",
 "laminate": {"catalog": "L001", "layup": "[0/0/0]"},
 "geometry": {"L": 1000, "section": {"type": "arc", "R": 20, "b": 62.83185307}, "size": 2.5},
 "element": {"type": "ASDShellQ4", "corotational": false},
 "supports": [{"edge": "x0", "dofs": [1, 1, 1, 1, 1, 1]}, {"rigid_end": "xL"}],
 "loads": [{"node": {"rigid": "xL"}, "force": [10, 0, 0]}],
 "analysis": {"type": "linear"},
 "report_node": {"rigid": "xL"}
}
EOF
cat > ts_L001_opp_bending.json <<'EOF'
{
 "name": "ts_L001_opp_bending",
 "laminate": {"catalog": "L001", "layup": "[0/0/0]"},
[... shortened ...]
```

```
usage: laminate_properties.py [-h] [--list-catalog] [--catalog ID] [--E1 E1]
                              [--E2 E2] [--nu12 NU12] [--G12 G12]
                              [--nu23 NU23] [--Xt XT] [--Xc XC] [--Yt YT]
                              [--Yc YC] [--S S] [--rho RHO] [--layup LAYUP]
                              [--angles ANGLES [ANGLES ...]]
                              [--thickness THICKNESS [THICKNESS ...]]
                              [--n-sub N_SUB] [--no-merge] [--G-out G_OUT]
                              [--check] [--save-json FILE] [--save-abd FILE]
                              [--emit FILE] [--json FILE] [--emit-code FILE]

[... shortened ...]
```

Bash Build L001 [0/0/0] laminate and check ABD

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
$PY ../../.claude/skills/laminate-properties/laminate_properties.py --catalog L001 --layup "[0/0/0]" --check --save-json
L001_000.json 2>&1 | tail -40
```

```
Q11 = 159399, Q22 = 9000.75, Q12 = 2844.24, Q66 = 4670 MPa

ABD from classical lamination theory:
A [N/mm]  (N = A eps + B k)
               eps_xx       eps_yy       gam_xy
Nxx           16736.9      298.645            0
Nyy           298.645      945.079            0
Nxy                 0            0       490.35
B [N]
                 k_xx         k_yy         k_xy
[... shortened ...]
```

Claude · 11:25

Laminate checks out (OpenSees ABD matches CLT). Now assembling and running the axial and bending models with 16 threads.

Bash Assemble and run axial and bending OpenSeesPy scripts

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
for c in ts_L001_axial ts_L001_opp_bending; do echo "=== $c"; time $PY ../../.claude/skills/shell-assemble-
run/assemble_run.py $c.json 2>&1 | tail -15; done
```

```
=== ts_L001_axial
Using ASDShellQ4 - Developed by: Massimo Petracca, Guido Camata, ASDEA Software Technology
1000 x 62.8319 mm shell, arc R = 20 mm (180.0 deg); 400 x 25 = 10000 quads of 2.5 x 2.51 mm, 10426 nodes
10000 elements, 62562 DOFs; linear solved in 4.4 s
saved F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\results\ts_L001_axial.npz
report node 10427: u = [0.009569, 0.0, -2e-06, -0.0, 0.0, 0.0]
max |u| = 0.00956926 mm
laminate piece: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\ts_L001_axial_code\laminate_code.py
    mesh piece: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\ts_L001_axial_code\mesh_code.py
      bc piece: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\ts_L001_axial_code\bc_code.py
[... shortened ...]
```

Bash Run nonlinear and refined-mesh bending checks

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
sed -e 's/"ts_L001_opp_bending"/"ts_L001_opp_bending_NL"/' -e 's/"corotational": false/"corotational": true/' -e 's/{"type":
"linear"}/{"type": "load_control", "n_steps": 10, "max_load_factor": 1}/' ts_L001_opp_bending.json >
ts_L001_opp_bending_NL.json
sed -e 's/"ts_L001_opp_bending"/"ts_L001_opp_bending_h1p25"/' -e 's/"size": 2.5/"size": 1.25/' ts_L001_opp_bending.json >
ts_L001_opp_bending_h1p25.json
cat ts_L001_opp_bending_NL.json
for c in ts_L001_opp_bending_NL ts_L001_opp_bending_h1p25; do echo "=== $c"; $PY ../../.claude/skills/shell-assemble-
run/assemble_run.py $c.json 2>&1 | grep -v "piece:\|Using" | tail -8; done
```

```
{
 "name": "ts_L001_opp_bending_NL",
 "laminate": {"catalog": "L001", "layup": "[0/0/0]"},
 "geometry": {"L": 1000, "section": {"type": "arc", "R": 20, "b": 62.83185307}, "size": 2.5},
 "element": {"type": "ASDShellQ4", "corotational": true},
 "supports": [{"edge": "x0", "dofs": [1, 1, 1, 1, 1, 1]}, {"rigid_end": "xL"}],
 "loads": [{"node": {"rigid": "xL"}, "force": [0, 0, -1]}],
 "analysis": {"type": "load_control", "n_steps": 10, "max_load_factor": 1},
 "report_node": {"rigid": "xL"}
}
[... shortened ...]
```

update_plan task list

```
{"done": [2, 3], "start": 4}
```

```
Plan updated: step 4 of 6 is in progress.
```

Bash Run Hashin failure check on axial and bending results

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
for c in ts_L001_axial ts_L001_opp_bending; do echo "=== $c"; $PY ../../.claude/skills/failure-check/failure_check.py
results/$c.npz --top 5 2>&1 | tee results/${c}_hashin.txt; done
```

```
=== ts_L001_axial
S23 = Yc/2 (default)
web facets: 0 of 10000 (no web (no catalog boom given))
Hashin 2D check: ts_L001_axial  [[0/0/0] h = 0.105 mm], 10000 facets, Gauss points, 3 points through each ply
rank         F        R  facet        x        y        z ply  theta     at         s1         s2        t12  mode
   1 4.027e-07     1576   3200     1.25   -9.616    2.508   1      0 bottom      1.587     0.0187  3.392e-07  fibre tension
   2 4.027e-07     1576   6799    998.8    9.616    2.508   1      0 bottom      1.587     0.0187  3.392e-07  fibre tension
   3 4.027e-07     1576   3599    998.8   -9.616    2.508   1      0 bottom      1.587     0.0187 -3.392e-07  fibre tension
   4 4.027e-07     1576   6400     1.25    9.616    2.508   1      0 bottom      1.587     0.0187 -3.392e-07  fibre tension
   5 4.027e-07     1576   4000     1.25   -4.964   0.6666   1      0 bottom      1.587     0.0187 -7.827e-07  fibre tension
[... shortened ...]
```

Bash Inspect failure_check function signatures

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills/failure-check"; grep -n "^def
\|^import\|^from\|sys.path\|^MODES\|^Z_POINTS\|^STRENGTHS" failure_check.py; sed -n "$(grep -n '^def failure_index'
failure_check.py | cut -d: -f1),+12p" failure_check.py
```

```
67:import argparse
68:import math
69:import sys
70:from pathlib import Path
72:import numpy as np
76:    sys.path.insert(0, str(_SKILLS / _d))
77:import laminate_properties as lp   # noqa: E402
79:STRENGTHS = lp.STRENGTHS
80:Z_POINTS = ('bottom', 'middle', 'top')
81:MODES = ('fibre tension', 'fibre compression', 'matrix tension', 'matrix compression')
[... shortened ...]
```

Write F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\coil_hub_check.py

```
"""
Hashin 2D check of a tape-spring flange coiled around a hub (2026-11-11, Tara).

Uses the failure-check skill's own functions (failure_check.coiling_check for
flattening only, failure_check.failure_index for flattening + hub bending).
failure_check.py's CLI only flattens the flange (k = [0, 1/r, 0]); coiling on a
hub of radius R_hub adds the longitudinal curvature k_xx = +-1/R_hub
(DOMAIN_KNOWLEDGE.md section 4). With N = 0, eps0 = -A^-1 B k (0 for a
symmetric layup). All four sign combinations (equal/opposite-sense coiling,
either face out) are checked; the largest F governs.

Usage: python coil_hub_check.py lam.json --r 20 --hub 40 [--midsurface]
[... shortened ...]
```

```
File created successfully at: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\coil_hub_check.py
(file state is current in your context — no need to Read it back)
```

Bash Run hub coiling Hashin check; peek NL run

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
$PY coil_hub_check.py L001_000.json --r 20 --hub 40 2>&1 | tee results/coil_hub_40mm.txt; tail -3
"C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25/tasks/baef7qof8.output"
```

```
Coiling check: flange [[0/0/0] h = 0.105 mm] flattened from r = 20 mm (tape-spring arc radius): k = +0.05 1/mm transverse
(k_xx = k_xy = 0, N = 0), eps0 = [3.52e-22, -9.41e-20, -0]
  max F = 0.1363 (matrix tension), R = 2.709, ply 3 (0 deg), top: s1 7.466, s2 23.63, t12 0 MPa
  ply  1 (     0 deg, z -0.0525..-0.0175): max F 0.006839
  ply  2 (     0 deg, z -0.0175..+0.0175): max F 0.01514
```

```
  ply  3 (     0 deg, z +0.0175..+0.0525): max F 0.1363

Coiling on a hub: R_hub = 40 mm, coil radius used 40 mm; flange r = 20 mm
  k = [+0.0250, +0.0500, 0] 1/mm (opposite-sense combination): max F = 0.1828 (matrix tension), R = 2.339, ply 3 top: s1
216.7, s2 27.36, t12 0 MPa
  k = [+0.0250, -0.0500, 0] 1/mm (equal-sense combination): max F = 0.09662 (matrix tension), R = 3.217, ply 1 bottom: s1
-201.7, s2 19.89, t12 0 MPa
  k = [-0.0250, +0.0500, 0] 1/mm (equal-sense combination): max F = 0.09662 (matrix tension), R = 3.217, ply 3 top: s1
-201.7, s2 19.89, t12 0 MPa
[... shortened ...]
```

Bash Compute boom mass with objective-constraints skill

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
$PY ../../.claude/skills/objective-constraints/objective_constraints.py results/ts_L001_opp_bending.npz --con "tip
deflection report: abs(tip_uz) <= 1e9" 2>&1 | tee results/ts_L001_opp_bending_objcon.txt | head -40
```

```
Objective / constraints: ts_L001_opp_bending  [[0/0/0] h = 0.105 mm]
OBJECTIVE  minimize mass: 0.0104169 kg (10.42 g)
CONSTRAINTS (g <= 0 is satisfied; g_rel = g / |limit|)
  [satisfied    ] no failure: F_governing < 1.0  value 0.136288, limit 1, g -0.863712, g_rel -0.863712
  [satisfied    ] tip deflection report: abs(tip_uz) <= 1000000000.0  value 8.52305, limit 1e+09, g -1e+09, g_rel -1
FEASIBLE: yes
  note: catalog L001 rho = 1580 kg/m^3
  note: tip = free end xL: rigid end master node at xL
  note: S23 = Yc/2 (default)
  note: coiling: flange flattened from r = 20 mm (smallest arc radius of the case cross-section), F = 0.1363 (matrix
tension)
[... shortened ...]
```

Claude · 11:36

Mass is 10.42 g (hand check: 1580 kg/m³ × 62.83 mm × 0.105 mm × 1 m = 10.42 g ✓). Waiting on the nonlinear bending cross-check.

update_plan task list

```
{"done": [4, 5], "start": 6}
```

```
Plan updated: step 6 of 6 is in progress. App actions available now: commit, create_pr, verify.
```

Bash Check nonlinear run progress

```
cat "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25/tasks/baef7qof8.output" | tail -5; ls -la "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26
(Tara)/results"; tasklist | grep -i python
```

```
"loads": [{"node": {"rigid": "xL"}, "force": [0, 0, -1]}],
 "analysis": {"type": "load_control", "n_steps": 10, "max_load_factor": 1},
 "report_node": {"rigid": "xL"}
}
=== ts_L001_opp_bending_NL
total 4564
drwxr-xr-x 1 tvenkata 1049089       0 Nov 11 11:36 .
drwxr-xr-x 1 tvenkata 1049089       0 Nov 11 11:36 ..
-rw-r--r-- 1 tvenkata 1049089    1323 Nov 11 11:36 coil_hub_40mm.txt
-rw-r--r-- 1 tvenkata 1049089 2297136 Nov 11 11:25 ts_L001_axial.npz
[... shortened ...]
```

ToolSearch

```
{
 "query": "select:TaskStop",
 "max_results": 1
}
```

```
[image]
```

TaskStop

```
{
 "task_id": "baef7qof8"
```

```
}
```

```
{"message":"Successfully stopped task: baef7qof8 (cd \"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26
(Tara)\"\nPY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe\nexport OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16
MKL_NUM_THREADS=16\nsed -e 's/\"ts_L001_opp_bending\"/\"ts_L001_opp_bending_NL\"/' -e 's/\"corotational\":
false/\"corotational\": true/' -e 's/{\"type\": \"linear\"}/{\"type\": \"load_control\", \"n_steps\": 10,
\"max_load_factor\": 1}/' ts_L001_opp_bending.json > ts_L001_opp_bending_NL.json\nsed -e
's/\"ts_L001_opp_bending\"/\"ts_L001_opp_bending_h1p25\"/' -e 's/\"size\": 2.5/\"size\": 1.25/' ts_L001_opp_bending.json >
ts_L001_opp_bending_h1p25.json\ncat ts_L001_opp_bending_NL.json\nfor c in ts_L001_opp_bending_NL ts_L001_opp_bending_h1p25;
do echo \"=== $c\"; $PY ../../.claude/skills/shell-assemble-run/assemble_run.py $c.json 2>&1 | grep -v \"piece:\\|Using\" |
tail -8; done)","task_id":"baef7qof8","task_type":"local_bash","command":"cd \"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code
for Project/11-11-26 (Tara)\"\nPY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe\nexport OMP_NUM_THREADS=16
OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16\nsed -e 's
[... shortened ...]
```

Bash Check whether the stopped run's Python is still alive

```
sleep 2; tasklist //FI "PID eq 50832" ; wmic process where "ProcessId=50832" get CommandLine 2>/dev/null | head -3
```

```
INFO: No tasks are running which match the specified criteria.
```

Claude · 11:37

Stopped cleanly. Running the refined-mesh linear check and a coarser (5 mm) nonlinear check instead, with live output.

Bash Run 1.25 mm linear and 5 mm nonlinear bending checks

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
sed -i 's/"size": 2.5/"size": 5/' ts_L001_opp_bending_NL.json
$PY ../../.claude/skills/shell-assemble-run/assemble_run.py ts_L001_opp_bending_h1p25.json 2>&1 | grep -v "piece:\|Using" |
tail -6
echo "=== NL 5 mm"
timeout 900 $PY -u ../../.claude/skills/shell-assemble-run/assemble_run.py ts_L001_opp_bending_NL.json 2>&1 | grep -v
"piece:\|Using" | tail -15
```

```
Command running in background with ID: b6i1v81p0. Output is being written to:
C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\4cfc3c23-d9ff-4995-bdcb-9adeea99fe25\tasks\b6i1v81p0.output.
You will be notified when it completes — do not poll that file or sleep while waiting. Read the file only if the user asks
for progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
Session cwd remains F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara); directory changes made by the
backgrounded command do not apply to subsequent commands.
```

Claude · 11:37

While that runs — beam-theory check of the linear tip deflection: $E_x = (A11 - A12^2/A22)/h$ = 158.5 GPa, $I_{arc} = R^3t(\pi/2 - 4/\pi)$ = 250 mm^4 → $PL^3/3EI$ = 8.41 mm vs FE 8.52 mm (+1.3%).

Bash Check progress of background bending checks

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; ls results; cat
"C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25/tasks/b6i1v81p0.output"
```

```
coil_hub_40mm.txt
ts_L001_axial.npz
ts_L001_axial_hashin.txt
ts_L001_opp_bending.npz
ts_L001_opp_bending_hashin.txt
ts_L001_opp_bending_objcon.json
ts_L001_opp_bending_objcon.txt
```

Claude · 11:37

Status update (~25 min): all three load cases, the Hashin checks and the mass are done. Two cross-checks of the bending case are still running in the background: a finer 1.25 mm mesh and a nonlinear run on a 5 mm mesh. I'll report everything when they finish.

[11:42] *Background command "Run 1.25 mm linear and 5 mm nonlinear bending checks" completed (exit code 0)*

Bash Read background run output

```
cat "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25/tasks/b6i1v81p0.output"
```

```
40000 elements, 245112 DOFs; linear solved in 18.8 s
saved F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\results\ts_L001_opp_bending_h1p25.npz
report node 40852: u = [0.0, -0.0, -8.502589, 0.0, 0.012631, 0.0]
max |u| = 8.50411 mm
joined script: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\ts_L001_opp_bending_h1p25_opensees.py (81150 lines)
ts_L001_opp_bending_h1p25_opensees.py ran in 31.0 s
=== NL 5 mm
NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()
StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 0.253125
OpenSees > analyze failed, returned: -3 error flag
[... shortened ...]
```

Bash Write displacement-control case; read solver

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
sed -e 's/ts_L001_opp_bending_NL/ts_L001_opp_bending_DC/' -e 's/{"type": "load_control", "n_steps": 10, "max_load_factor":
1}/{"type": "displacement_control", "node": {"rigid": "xL"}, "dof": 3, "step": -0.5, "n_steps": 80, "max_load_factor": 1.0,
"tol": 1e-6}/' ts_L001_opp_bending_NL.json > ts_L001_opp_bending_DC.json; cat ts_L001_opp_bending_DC.json; grep -n "def
displacement_control" -A30 ../../.claude/skills/shell-bc-load/shell_bc_load.py | head -45
```

```
{
 "name": "ts_L001_opp_bending_DC",
 "laminate": {"catalog": "L001", "layup": "[0/0/0]"},
 "geometry": {"L": 1000, "section": {"type": "arc", "R": 20, "b": 62.83185307}, "size": 5},
 "element": {"type": "ASDShellQ4", "corotational": true},
 "supports": [{"edge": "x0", "dofs": [1, 1, 1, 1, 1, 1]}, {"rigid_end": "xL"}],
 "loads": [{"node": {"rigid": "xL"}, "force": [0, 0, -1]}],
 "analysis": {"type": "displacement_control", "node": {"rigid": "xL"}, "dof": 3, "step": -0.5, "n_steps": 80,
"max_load_factor": 1.0, "tol": 1e-6},
 "report_node": {"rigid": "xL"}
}
[... shortened ...]
```

Bash Run displacement-control opposite-sense bending

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
$PY -u ../../.claude/skills/shell-assemble-run/assemble_run.py ts_L001_opp_bending_DC.json 2>&1 | grep -v
"piece:\|Using\|WARNING\|NewtonRaphson\|StaticAnalysis\|OpenSees >\|after:"
```

```
Command running in background with ID: burghyaat. Output is being written to:
C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\4cfc3c23-d9ff-4995-bdcb-9adeea99fe25\tasks\burghyaat.output.
You will be notified when it completes — do not poll that file or sleep while waiting. Read the file only if the user asks
for progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
Session cwd remains F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara); directory changes made by the
backgrounded command do not apply to subsequent commands.
```

Claude · 11:42

The displacement-control run is going in the background; I'll report once it finishes.

[11:47] *Background command "Run displacement-control opposite-sense bending" completed (exit code 0)*

Bash Read displacement-control output

```
cat "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25/tasks/burghyaat.output" | tail -40
```

```
joined script: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\ts_L001_opp_bending_DC_opensees.py
(5651 lines)
  step   1: load factor 0.05816, 7 iterations
  step   2: load factor 0.11628, 10 iterations
  step   3: load factor 0.17434, 14 iterations
  step   4: load factor 0.2323, 38 iterations
```

```
  stopped: no convergence beyond load factor 0.2323
1000 x 62.8319 mm shell, arc R = 20 mm (180.0 deg); 200 x 13 = 2600 quads of 5 x 4.83 mm, 2814 nodes
2600 elements, 16890 DOFs; displacement_control solved in 304.5 s
saved F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\results\ts_L001_opp_bending_DC.npz
report node 2815: u = [-0.002387, 0.0, -2.0, -0.0, 0.002972, 0.0]
[... shortened ...]
```

Bash Read shell-buckling skill docs

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills/shell-buckling"; sed -n 1,120p SKILL.md
```

```
---
name: shell-buckling
description: Buckling strength of thin-shell booms (tape springs, TRAC, omega, CTM or any shell-mesh section) under
compression, bending or any load combination in OpenSeesPy - estimate and buckling mode from the zero crossing of the lowest
tangent-stiffness eigenvalue along the nonlinear path (ShellNLDKGQ; the OpenSees counterpart of the Abaqus "preload +
eigenvalue" procedure), then the buckling strength from imperfection-seeded path following with ASDShellQ4 (Riks-type), with
thin-walled section properties and Euler / flexural-torsional column formulas for checks. Use whenever a buckling load,
buckling moment or buckling mode is asked for.
---

# shell-buckling

Scripts (this folder; import `../laminate-properties`, `../shell-mesh`, `../shell-bc-load`):
`shell_buckling.py` (library), `boom_buckling.py` (case-file CLI), `examples/*.json`.
Verification and study: `Code for Project/10-29-26 (Federico)/` (`BUCKLING_STUDY.md`,
[... shortened ...]
```

Bash Run opposite-sense moment buckling with shell-buckling skill

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
cat > ts_L001_opp_moment_buckling.json <<'EOF'
{"name": "ts_L001_opp_moment_buckling",
 "laminate": {"catalog": "L001", "layup": "[0/0/0]"},
 "geometry": {"L": 200, "section": {"type": "arc", "R": 20, "b": 62.83185307}, "size": 2.5},
 "load": {"type": "bending", "axis": "y", "sign": 1},
 "A": {"step": 0.002, "max": 0.1, "n_coarse": 8},
 "B": {"step": 0.002, "max": 0.1, "amplitude": 0.0105, "control": "disp"}}
EOF
$PY -u ../../.claude/skills/shell-buckling/boom_buckling.py ts_L001_opp_moment_buckling.json --out results 2>&1 | grep -v
"WARNING\|NewtonRaphson\|StaticAnalysis\|OpenSees >\|after:\|Using"
```

```
Command running in background with ID: bkk1zj000. Output is being written to:
C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\4cfc3c23-d9ff-4995-bdcb-9adeea99fe25\tasks\bkk1zj000.output.
You will be notified when it completes — do not poll that file or sleep while waiting. Read the file only if the user asks
for progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
Session cwd remains F:\Tara\BenazzoVenkatadri_ThinShellSims\.claude\skills\shell-buckling; directory changes made by the
backgrounded command do not apply to subsequent commands.
```

Claude · 11:48

The shell-buckling run (200 mm segment under uniform opposite-sense moment, 2.5 mm mesh) is running in the background. Its peak moment will show whether the 1000 N·mm root moment from a 1 N tip load is below the snap moment.

User · 13:02

Status update please

Bash Read buckling run progress

```
tail -25 "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25/tasks/bkk1zj000.output"
```

```
(Bash completed with no output)
```

Claude · 13:03

Status (~50 min in):

Done - **Axial 10 N:** tip extension 0.0096 mm, within 0.6% of PL/EA. Hashin max F = 4.0×10^{-7} (fibre tension). - **Opposite-sense tip load 1 N, linear:** tip deflection **8.52 mm**. The 1.25 mm mesh gives 8.50 mm, and beam theory gives 8.41 mm. Hashin max F = 4.2×10^{-4} (fibre tension, s1 = 50 MPa at the root edges). - **Coiling on a 40 mm hub:** Hashin F = **0.183** (matrix tension, strength ratio 2.34). Flattening alone gives F = 0.136. - **Mass: 10.42 g**.

Still running - A shell-buckling check of the opposite-sense peak moment. It has printed nothing yet; the grep filter is probably buffering its output.

Why it's running: two nonlinear tip-load runs (load control, then displacement control) both stopped converging at about 0.23–0.25 N. The path was still linear there, with no softening. The skill docs say the rigid-end link used by shell-bc-load is unreliable under large rotations, so this is probably a solver problem rather than a real snap. The buckling run will confirm whether the 1000 N·mm root moment from the 1 N load stays below the snap moment. Once it finishes I'll send the final report.

User · 13:04

Do you expect this bending problem to be beyond the linear regime?

Bash Compare transverse curvature change to 1/R in linear result

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
/c/Users/tvenkata/miniforge3/envs/shells/python.exe -c "
import sys; sys.path.insert(0,'../../.claude/skills/shell-bc-load'); sys.path.insert(0,'../../.claude/skills/shell-mesh');
sys.path.insert(0,'../../.claude/skills/laminate-properties')
import shell_bc_load as bc, numpy as np
r=bc.load_results('results/ts_L001_opp_bending.npz'); E=r['strains']; c=r['points'][r['faces']].mean(1)
kxx,kyy=E[...,3].mean(1),E[...,4].mean(1)
print('max|kyy| =',abs(kyy).max(),'1/mm  vs 1/R =',1/20,' ratio',abs(kyy).max()*20)
m=(c[:,0]>50)&(c[:,0]<950); print('away from ends: max|kyy| =',abs(kyy[m]).max(),' ratio',abs(kyy[m]).max()*20)
print('max|kxx| =',abs(kxx).max(),' M/EI root =',1000/3.962e7)
"
```

```
max|kyy| = 0.00010311473544161412 1/mm  vs 1/R = 0.05  ratio 0.0020622947088322825
away from ends: max|kyy| = 6.164231307023149e-05  ratio 0.0012328462614046298
max|kxx| = 0.00023911194698719647  M/EI root = 2.523977788995457e-05
```

Claude · 13:04

Not in the usual beam sense, but the cross-section might flatten enough that a linear analysis is unreliable at 1 N. I won't know until the buckling run finishes.

Beam level: clearly linear. The load parameter $PL^2/EI = (1 \times 1000^2)/(3.96\times10^7) = 0.025$, far below the ~1 where large-rotation effects start to matter. The tip deflection is 8.5 mm on a 1 m boom (0.85% of L).

Cross-section level: uncertain. Opposite-sense bending of a tape spring fails when the curved cross-section flattens and then snaps. A linear analysis cannot capture this, because the flattening grows with the square of the bending curvature. - In the linear run, the transverse curvature change is only 0.1–0.2% of 1/R, but that is the linear part only. - A Brazier-type estimate of the flattening uses the parameter $(E1/E2)(\kappa R^2/t)^2$. - $\kappa = M/EI = 2.5\times10^{-5}$ /mm at the root. - For an isotropic tape this gives about 0.009. - The [0/0/0] layup multiplies it by $E1/E2 \approx 17.7$, giving about 0.16. - For a closed tube the peak moment comes at about 0.22, and an open arc flattens more easily than a tube. So the 1000 N·mm root moment may not be far from the snap moment of this very anisotropic tape. Its transverse bending stiffness D22 is only 0.87 N·mm, against D11 = 15.4. - The fold moment, $(1+\nu)D\alpha \approx 12$ N·mm, is the moment the tape carries after it snaps, not the snap moment. It doesn't tell us whether 1000 N·mm is safe.

The two stalled nonlinear runs. Both stopped near 0.25 N, a root moment of about 250 N·mm. At that moment the flattening parameter is only about 0.01, and the load–deflection path was still exactly linear. So I

still think those stalls are the rigid-end problem described in the skill docs, not a physical snap.

The shell-buckling run under opposite-sense moment will give the peak moment directly: - If the peak is well above 1000 N·mm, the linear results (8.5 mm tip deflection, F = 4×10^{-4}) stand. - If it is near or below 1000 N·mm, a 1 N tip load would snap the boom near the root, and the linear results would not be valid.

User · 13:05

How much longer is the analysis likely to take?

Bash Check buckling run progress and outputs

```
cat "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-9adeea99fe25/tasks/bkk1zj000.output" | tail; ls -la "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)/results" | tail -5; grep -n "print" "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills/shell-buckling/boom_buckling.py" | head
```

```
-rw-r--r-- 1 tvenkata 1049089  631955 Nov 11 11:42 ts_L001_opp_bending_NL.npz
-rw-r--r-- 1 tvenkata 1049089 9169105 Nov 11 11:37 ts_L001_opp_bending_h1p25.npz
-rw-r--r-- 1 tvenkata 1049089    2118 Nov 11 11:36 ts_L001_opp_bending_hashin.txt
-rw-r--r-- 1 tvenkata 1049089    7892 Nov 11 11:36 ts_L001_opp_bending_objcon.json
-rw-r--r-- 1 tvenkata 1049089     948 Nov 11 11:36 ts_L001_opp_bending_objcon.txt
187:            print(f'A (ShellNLDKGQ stability scan): STABLE to {res["A"]["critical_load"]} at '
191:            print(f'A (ShellNLDKGQ stability scan): critical {res["A"]["critical_load"]} at '
194:        print('A skipped: off-axis plies (ShellNLDKGQ takes 0/90 only)')
225:        print(f'B (ASDShellQ4 imperfect path, amplitude {amp:g} mm): peak {peak:.5g} at '
230:    print(f'saved {out_dir / (name + ".json")}')
```

PowerShell Show running Python processes start time and CPU

```
Get-Process python -ErrorAction SilentlyContinue | Where-Object { $_.SessionId -eq (Get-Process -Id $PID).SessionId } | Select-Object Id, StartTime, @{n='CPU_s';e={[int]$_.CPU}}, @{n='MB';e={[int]($_.WS/1MB)}} | Format-Table; Get-Date
```

```
Id StartTime              CPU_s  MB
    -- ---------              -----  --
 27032 11/10/2026 5:50:37 PM      1  50
 76984 11/11/2026 11:48:10 AM  4717 205
116164 11/10/2026 5:50:39 PM      1  36
138316 11/10/2026 5:47:50 PM      1  36



Wednesday, November 11, 2026 1:06:04 PM
```

Bash Check how --only B gets its imperfection

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills/shell-buckling"; grep -n "only\|amplitude\|mode" boom_buckling.py | sed -n 1,40p
```

```
4:    python boom_buckling.py case.json [--out DIR] [--only A|B]
9:     buckling mode. Estimate only: exact for flat plates and out-of-plane
10:     (local) modes, too high for global modes of curved sections (tape
14:     any laminate): the mode of A (or a mode file) scaled to the imperfection
15:     amplitude, then rotation/end-shortening control (or load control); the
27: "B": {"step": 0.005, "max": 0.3, "amplitude": 0.01, "control": "disp"},   # or "load" (step, max = load factor)
28: "mode_file": null
32:2020 style); about z: free in ux, uy, rz. Compression: tip free in ux only
65:def build_case(case, element, imperfection=None, amplitude=0.0):
70:    sb.build_shell(mesh, lam, element, imperfection, amplitude)
[... shortened ...]
```

wrap_up wrap-up

```
The axial, linear bending and hub-coiling cases, their Hashin checks, and the mass (10.42 g) are done. The buckling check of the opposite-sense peak moment has been running for 78 min, is still in its first phase (A), and has printed nothing.
```

```
Shown in the next-step box. Your closing message is your answer to the user: write it now, in full, as the last thing you say this turn. The box asks your question, so do not ask it again.
```

Claude · 13:06

I can't give a reliable estimate, and it's already taking much longer than I expected.

- **Where it is:** the buckling job started at 11:48 and has run for 78 minutes (it's 13:06 now). It has used about 4,700 CPU-seconds and is still in phase A, a scan that only estimates the buckling moment and mode. Phase A rebuilds the model from scratch for every evaluation, with up to 50 rotation steps each, and it doesn't report progress. It could need anywhere from a few more minutes to another hour.
- **Phase B is what we need:** it follows the bending path with rotation control, and its peak moment is what decides whether 1000 N·mm is safe. On this 12k-DOF model, B by itself should take roughly 5–15 minutes.
- **Faster option:** stop the job and run B by itself. Without phase A there's no buckling mode to seed an imperfection, so B runs on the perfect tape. That's acceptable here, because opposite-sense bending fails at a snap the perfect model can still find. The risk is a slightly high peak moment; I'll say so in the report.

Everything else is finished, in `Code for Project/11-11-26 (Tara)/`:

Case	Result	Hashin max F	Mode
Axial, 10 N	tip extension 0.0096 mm	4.0×10^{-7}	fibre tension
Tip load 1 N, opposite-sense (linear)	tip deflection **8.52 mm**	4.2×10^{-4}	fibre tension at root edges
Coiling on 40 mm hub (flatten + bend)	strength ratio 2.34	**0.183**	matrix tension

The boom mass is **10.42 g**.

The bending result depends on the peak-moment check: if the snap moment is near or below 1000 N·mm, the linear tip deflection and F for the 1 N case are not valid.

User · 13:07

Do just part B for now

update_plan task list

```
[in_progress] Stop phase A buckling scan
[pending] Run phase B peak-moment path
[pending] Report final results
```

TaskStop

```
{
 "task_id": "bkk1zj000"
}
```

```
Plan shown (3 steps).
```


```
{"message":"Successfully stopped task: bkk1zj000 (cd \"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26
(Tara)\"\nPY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe\nexport OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16
MKL_NUM_THREADS=16\ncat > ts_L001_opp_moment_buckling.json <<'EOF'\n{\"name\": \"ts_L001_opp_moment_buckling\",\n
\"laminate\": {\"catalog\": \"L001\", \"layup\": \"[0/0/0]\"},\n \"geometry\": {\"L\": 200, \"section\": {\"type\": \"arc\",
\"R\": 20, \"b\": 62.83185307}, \"size\": 2.5},\n \"load\": {\"type\": \"bending\", \"axis\": \"y\", \"sign\": 1},\n \"A\":
{\"step\": 0.002, \"max\": 0.1, \"n_coarse\": 8},\n \"B\": {\"step\": 0.002, \"max\": 0.1, \"amplitude\": 0.0105,
\"control\": \"disp\"}}\nEOF\n$PY -u ../../.claude/skills/shell-buckling/boom_buckling.py ts_L001_opp_moment_buckling.json -
-out results 2>&1 | grep -v \"WARNING\\|NewtonRaphson\\|StaticAnalysis\\|OpenSees
>\\|after:\\|Using\")","task_id":"bkk1zj000","task_type":"local_bash","command":"cd
\"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26
(Tara)\"\nPY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe\nexport OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16
MKL_NUM_THRE
[... shortened ...]
```


PowerShell Make sure the phase A Python process is gone

```
Start-Sleep 2; $p = Get-Process -Id 76984 -ErrorAction SilentlyContinue; if ($p) { Stop-Process -Id 76984 -Force -
Confirm:$false; "killed 76984" } else { "76984 already gone" }
```

```
76984 already gone
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 2 of 3 is in progress.
```

Bash Run phase B opposite-sense peak-moment path only

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
$PY -u ../../.claude/skills/shell-buckling/boom_buckling.py ts_L001_opp_moment_buckling.json --out results --only B >
results/ts_L001_opp_moment_buckling_B.log 2>&1
grep -v "WARNING\|NewtonRaphson\|StaticAnalysis\|OpenSees >\|after:\|Using" results/ts_L001_opp_moment_buckling_B.log | tail
-30
```

```
Command running in background with ID: b5m3q8dqx. Output is being written to:
C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\4cfc3c23-d9ff-4995-bdcb-9adeea99fe25\tasks\b5m3q8dqx.output.
You will be notified when it completes — do not poll that file or sleep while waiting. Read the file only if the user asks
for progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
Session cwd remains F:\Tara\BenazzoVenkatadri_ThinShellSims\.claude\skills\shell-buckling; directory changes made by the
backgrounded command do not apply to subsequent commands.
```

Claude · 13:08

I stopped phase A (its process has exited). Phase B is now running on its own: a perfect tape under opposite-sense moment, rotation control, 2.5 mm mesh. I expect it to take about 5–15 minutes. Its step-by-step progress goes to `results/ts_L001_opp_moment_buckling_B.log`, so you can follow it. I'll report the peak moment as soon as it finishes.

User · 13:13

How is it going?

Bash Check phase B log progress

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; date; ls -la
results/ts_L001_opp_moment_buckling_B.log; grep -v "WARNING\|NewtonRaphson\|StaticAnalysis\|OpenSees >\|after:\|Using"
results/ts_L001_opp_moment_buckling_B.log | tail -15; grep -c "failed to converge" results/ts_L001_opp_moment_buckling_B.log
```

```
Wed Nov 11 13:13:10 PST 2026
-rw-r--r-- 1 tvenkata 1049089 990 Nov 11 13:13 results/ts_L001_opp_moment_buckling_B.log
  step   1: param  0.002, load factor  394.879, iters 12
  step   2: param  0.004, load factor  789.478, iters 16
  step   3: param  0.006, load factor  1183.53, iters 23
  step   4: param  0.008, load factor  1576.79, iters 28
  step   5: param  0.01, load factor  1968.98, iters 35
  step   6: param  0.012, load factor  2359.8, iters 43
  step   7: param  0.014, load factor  2748.89, iters 56
  step   8: param  0.016, load factor  3135.77, iters 67
[... shortened ...]
```

Claude · 13:13

It's going well, and it has already answered the key question.

The path has reached a rotation of 0.017 rad. The moment is **3,328 N·mm and still rising**, at about 3.3 times the 1,000 N·mm root moment from the 1 N tip load. So the boom does not snap under the 1 N opposite-sense load. The 1,000 N·mm level was passed between steps 2 and 3, and up to there the path is linear. The stiffness at that point is 197,000 N·mm/rad, which matches EI/L = 3.96×10^{7} / 200 = 198,000 N·mm/rad within 0.5%.

This means the linear results for the 1 N case stand: **tip deflection 8.52 mm** and Hashin F = 4.2×10^{-4}. It also confirms the two earlier nonlinear tip-load stalls near 0.25 N were solver failures, not a real snap.

The run is slowing down: Newton iterations per step have grown from 12 to 87, and one step needed a cut. That usually means the peak is getting close. I'll let it continue to find the actual snap moment, which gives the safety margin against snap, and report when it finishes or stalls. If it stalls while the moment is still rising, the result is only a lower bound of about 3,300 N·mm or more. That is still a margin of at least 3.3.

User · 13:14

If it has answered the question already and ensured that there is no buckling under the desired load, I don't need to know what the buckling moment is

TaskStop

```
{
 "task_id": "b5m3q8dqx"
}
```


```
{"message":"Successfully stopped task: b5m3q8dqx (cd \"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)\"\nPY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe\nexport OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16\n$PY -u ../../.claude/skills/shell-buckling/boom_buckling.py ts_L001_opp_moment_buckling.json --out results --only B > results/ts_L001_opp_moment_buckling_B.log 2>&1\ngrep -v \"WARNING\\|NewtonRaphson\\|StaticAnalysis\\|OpenSees >\\|after:\\|Using\" results/ts_L001_opp_moment_buckling_B.log | tail -30)","task_id":"b5m3q8dqx","task_type":"local_bash","command":"cd \"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)\"\nPY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe\nexport OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16\n$PY -u ../../.claude/skills/shell-buckling/boom_buckling.py ts_L001_opp_moment_buckling.json --out results --only B > results/ts_L001_opp_moment_buckling_B.log 2>&1\ngrep -v \"WARNING\\|NewtonRaphson\\|StaticAnalysis\\|OpenSees >\\|after:\\|Using\" results/ts_L001_opp_moment_buckling_B.log | tail -30"}
```


PowerShell Ensure no buckling Python process remains

```
Start-Sleep 2; Get-CimInstance Win32_Process -Filter "Name='python.exe'" | Where-Object { $_.CommandLine -like '*boom_buckling*' } | ForEach-Object { Stop-Process -Id $_.ProcessId -Force -Confirm:$false; "killed $($_.ProcessId)" }; "check done"
```

```
check done
```

Write F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\SUMMARY.md

```
# 11-11-26 (Tara): tape spring L001 [0/0/0], R = 20 mm, 180 deg, L = 1000 mm, cantilever

Laminate: L001 (IM7/8552), [0/0/0], t_ply 0.035 mm, h = 0.105 mm (`L001_000.json`; OpenSees ABD vs CLT: A, B exact, D -0.4%).
Section: arc R = 20 mm, b = pi R = 62.83 mm (centre of curvature +z, edges up). Root edge x0 clamped, tip xL rigid end.
Mesh: ASDShellQ4, 2.5 mm (400 x 25 = 10000 elements, 62562 DOFs). Python: shells env, 16 BLAS threads.
Hashin 2D (failure-check skill), S23 = Yc/2 default; catalog mean RTD strengths.

| Case | Script / case | Result | Max Hashin F | Mode, location |
|---|---|---|---|---|
| Axial 10 N (linear) | `ts_L001_axial*.json/_opensees.py` | tip ux = 0.00957 mm (PL/EA 0.00951, +0.6%) | 4.0e-7 (R = 1576) | fibre tension, uniform s1 = 1.59 MPa |
| Tip load 1 N opposite sense, Fz = -1 (linear) | `ts_L001_opp_bending*` | **tip uz = -8.52 mm** (1.25 mm mesh: 8.50; PL^3/3EI = 8.41, EI = 3.96e7 N mm^2) | 4.2e-4 (R = 49) | fibre tension, s1 = 50 MPa at root edges (Mc/I = 51) |
| Coiling on 40 mm hub (flatten r = 20 + k_x = 1/40, worst of 4 signs) | `coil_hub_check.py` -> `results/coil_hub_40mm.txt` | strength ratio 2.34 | **0.183** | matrix tension, outer ply, s1 = 217,
[... shortened ...]
```

```
File created successfully at: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\SUMMARY.md (file state is current in your context — no need to Read it back)
```

Edit F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\SUMMARY.md

```
- = 1000 N mm < 3328/3.3, so no
+ = 1000 N mm, at least 3.3x below that, so no
```

wrap_up wrap-up

```
All load cases, the Hashin checks and the mass are done. The bending check showed no snap at the 1 N load, and the run was stopped by request. Code, results and SUMMARY.md are in Code for Project/11-11-26 (Tara); nothing is committed.
```

```
The file F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\SUMMARY.md has been updated successfully. (file state is current in your context — no need to Read it back)
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.
```

Claude · 13:15

I stopped the buckling run. The boom does not snap or buckle under the 1 N load: the moment was still rising at 3,328 N·mm, at least 3.3 times the 1,000 N·mm root moment, so the linear bending results stand. The snap moment itself was not found.

The boom is a tape spring: L001 (IM7/8552), [0/0/0], 0.105 mm thick, R = 20 mm, 180°, L = 1 m, clamped at the root. Hashin F is the failure index (failure at F ≥ 1), and the strength ratio is the factor on the load that would bring the boom to failure.

Load case	Result	Hashin max F	How close to failure
Axial, 10 N	tip extension 0.0096 mm	4.0×10^{-7}	strength ratio about 1,600 (fibre tension)
Tip load 1 N, opposite-sense	**tip deflection 8.52 mm**	4.2×10^{-4}	strength ratio about 49 (fibre tension at the root edges, 50 MPa)
Coiling on a 40 mm hub	—	**0.183**	strength ratio 2.34 (matrix tension in the outer ply)

Boom mass: 10.42 g. A hand check (density × width × thickness × length) gives the same value.

- **Coiling governs.** It is the only case that comes close to failure, but it is still well below F = 1.
- **The answers were checked:**
- The axial extension is within 0.6% of hand theory (PL/EA).
- The tip deflection is 1.3% above beam theory (8.41 mm), and a finer mesh changes it by only 0.2%.
- The bending stress matches the hand value (Mc/I).
- **The coiling check needed extra code.** The failure-check skill's built-in coiling check only flattens the cross-section; it doesn't include bending around the hub. So I wrote `coil_hub_check.py`, which uses the skill's own functions to add the hub curvature. It checks all four sign combinations and keeps the worst. It also uses the 40 mm hub radius instead of the slightly larger radius of the boom's mid-surface, which is a bit conservative. Flattening alone gives F = 0.136.
- **Strengths:** they are catalog mean values, not design allowables, and they were measured on thicker plies than the 35 µm plies used here. The matrix-compression strength S23 is the default Yc/2, because it isn't in the catalog.
- **Nonlinear runs of the 1 N tip load don't work as set up.** They stop converging at about 0.25 N even though the response is still linear. That points to the solver's rigid tip connection, not to the boom.

I limited the solver to 16 threads and wrote files only inside the project folder.

On status updates: I can't send messages on a timer, so I posted them as each step finished instead of every 15 minutes.

Everything is in `Code for Project/11-11-26 (Tara)/`: the case files, the assembled OpenSeesPy scripts, `coil_hub_check.py`, the `results/` folder and SUMMARY.md. Nothing is committed.

User · 13:19

What about a [90/90/90] layup under the same axial and bending loads? Would material failure occur?

update_plan task list

```
[in_progress] Write [90/90/90] axial and bending cases
[pending] Run both OpenSeesPy models
[pending] Hashin failure checks
[pending] Check nonlinearity of bending case
[pending] Report results
```

```
Plan shown (5 steps).
```

Bash Create and run [90/90/90] axial and bending models

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
for c in axial opp_bending; do sed -e "s/ts_L001_$c/ts_L001_90_$c/" -e 's#\[0/0/0\]#[90/90/90]#' ts_L001_$c.json >
ts_L001_90_$c.json; done
grep -h '"name"\|layup' ts_L001_90_*.json
$PY ../../.claude/skills/laminate-properties/laminate_properties.py --catalog L001 --layup "[90/90/90]" --check --save-json
L001_909090.json 2>&1 | grep -A4 "^D \[N mm\]" | head -5
for c in ts_L001_90_axial ts_L001_90_opp_bending; do echo "=== $c"; $PY ../../.claude/skills/shell-assemble-
run/assemble_run.py $c.json 2>&1 | grep -v "piece:\|Using" | tail -5; done
```

```
"name": "ts_L001_90_axial",
 "laminate": {"catalog": "L001", "layup": "[90/90/90]"},
 "name": "ts_L001_90_opp_bending",
 "laminate": {"catalog": "L001", "layup": "[90/90/90]"},
D [N mm]  (M = B eps + D k)
                 k_xx         k_yy         k_xy
Mxx          0.868291      0.27438            0
Myy           0.27438       15.377            0
Mxy                 0            0     0.450509
=== ts_L001_90_axial
[... shortened ...]
```

update_plan task list

```
{"done": [1, 2], "start": 3}
```

```
Plan updated: step 3 of 5 is in progress.
```

Bash Hashin checks for [90/90/90] cases

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
for c in ts_L001_90_axial ts_L001_90_opp_bending; do echo "=== $c"; $PY ../../.claude/skills/failure-check/failure_check.py
results/$c.npz --top 6 > results/${c}_hashin.txt 2>&1; sed -n '3,10p;/^MAX F/p;/^Per ply/,/ply  3/p;/GOVERNING/p'
results/${c}_hashin.txt; done
```

```
=== ts_L001_90_axial
Hashin 2D check: ts_L001_90_axial  [[90/90/90] h = 0.105 mm], 10000 facets, Gauss points, 3 points through each ply
rank         F        R  facet        x        y        z ply  theta     at         s1         s2        t12  mode
   1 0.0005764    41.65   5198    996.2 1.665e-15  0.03947   3     90    top   -0.04494      1.537  5.481e-09  matrix
tension
   2 0.0005764    41.65   5198    996.2 1.665e-15  0.03947   3     90    top   -0.04494      1.537 -5.479e-09  matrix
tension
   3 0.0005764    41.65   4801     3.75 1.665e-15  0.03947   3     90    top   -0.04494      1.537  5.474e-09  matrix
tension
   4 0.0005764    41.65   4801     3.75 1.665e-15  0.03947   3     90    top   -0.04494      1.537 -5.486e-09  matrix
tension
   5 0.0005764    41.65   5198    996.2 1.665e-15  0.03947   3     90    top   -0.04056      1.537  5.481e-09  matrix
tension
   6 0.0005764    41.65   5198    996.2 1.665e-15  0.03947   3     90    top   -0.04056      1.537 -5.479e-09  matrix
tension
Per ply (max over the model):
[... shortened ...]
```

Claude · 13:20

Linear results are in. Two checks before I trust them: a finer mesh, since the peak F is at the clamp corner, and whether the root moment stays below the snap moment for this much softer layup.

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 4 of 5 is in progress.
```

Bash Start snap check; run refined-mesh bending and Hashin

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
export OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16
sed -e 's/ts_L001_opp_moment_buckling/ts_L001_90_opp_moment_buckling/' -e 's#\[0/0/0\]#[90/90/90]#' -e 's/"step": 0.002,
```

```
"max": 0.1, "amplitude"/"step": 0.01, "max": 0.15, "amplitude"/' ts_L001_opp_moment_buckling.json >
ts_L001_90_opp_moment_buckling.json
cat ts_L001_90_opp_moment_buckling.json
nohup $PY -u ../../.claude/skills/shell-buckling/boom_buckling.py ts_L001_90_opp_moment_buckling.json --out results --only B
> results/ts_L001_90_opp_moment_buckling_B.log 2>&1 &
sed -e 's/ts_L001_90_opp_bending/ts_L001_90_opp_bending_h1p25/' -e 's/"size": 2.5/"size": 1.25/' ts_L001_90_opp_bending.json
> ts_L001_90_opp_bending_h1p25.json
$PY ../../.claude/skills/shell-assemble-run/assemble_run.py ts_L001_90_opp_bending_h1p25.json 2>&1 | grep "report node"
$PY ../../.claude/skills/failure-check/failure_check.py results/ts_L001_90_opp_bending_h1p25.npz --top 12 >
results/ts_L001_90_opp_bending_h1p25_hashin.txt 2>&1
sed -n '4,16p;/^MAX F/p' results/ts_L001_90_opp_bending_h1p25_hashin.txt
```

```
{"name": "ts_L001_90_opp_moment_buckling",
 "laminate": {"catalog": "L001", "layup": "[90/90/90]"},
 "geometry": {"L": 200, "section": {"type": "arc", "R": 20, "b": 62.83185307}, "size": 2.5},
 "load": {"type": "bending", "axis": "y", "sign": 1},
 "A": {"step": 0.002, "max": 0.1, "n_coarse": 8},
 "B": {"step": 0.01, "max": 0.15, "amplitude": 0.0105, "control": "disp"}}
report node 40852: u = [1e-06, -8.6e-05, -149.208377, 6e-06, 0.22369, -0.0]
rank         F        R  facet        x        y        z ply  theta     at         s1         s2        t12  mode
   1    0.6212    1.269      0    0.625   -19.98    19.37   1     90 bottom       11.5      50.44    -0.7721  matrix tension
   2    0.6212    1.269  39200    0.625    19.98    19.37   1     90 bottom       11.5      50.44     0.7721  matrix tension
[... shortened ...]
```

Bash Check snap-check progress

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; grep "step\|peak\|stopped"
results/ts_L001_90_opp_moment_buckling_B.log | tail
```

```
step   1: param  0.01, load factor  111.384, iters 11
  step   2: param  0.02, load factor  222.213, iters 14
  step   3: param  0.03, load factor  332.12, iters 16
  step   4: param  0.04, load factor  440.838, iters 17
```

ToolSearch

```
{
 "query": "select:Monitor",
 "max_results": 1
}
```

```
[image]
```

Bash Wait until snap check passes 1000 N mm or stops

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)";
L=results/ts_L001_90_opp_moment_buckling_B.log
until grep -qE "load factor +1[0-9]{3}\.|stopped|peak|Traceback|Error" $L; do sleep 5; done; grep
"step\|peak\|stopped\|Traceback\|Error" $L | tail -14
```

```
Command running in background with ID: biz0n8cwi. Output is being written to:
C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\4cfc3c23-d9ff-4995-bdcb-9adeea99fe25\tasks\biz0n8cwi.output.
You will be notified when it completes — do not poll that file or sleep while waiting. Read the file only if the user asks
for progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
Session cwd remains F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara); directory changes made by the
backgrounded command do not apply to subsequent commands.
```

Claude · 13:21

Preliminary answer: no material failure is predicted for either load case with [90/90/90]. The bending case comes much closer than with [0/0/0].

Case	Result	Hashin max F	Strength ratio
Axial, 10 N	tip extension 0.169 mm (hand calc 0.169)	5.8×10^{-4}	42
Tip load 1 N, opposite-sense (linear)	**tip deflection 149.6 mm** (beam theory 149.0)	**0.59** (1.25 mm mesh: 0.62)	1.3

> I'm still checking that the root bending moment stays below the snap moment for this layup, and will confirm when it reaches 1000 N·mm.

[13:22] *Background command "Wait until snap check passes 1000 N mm or stops" completed (exit code 0)*

Bash Read snap-check result

```
cat "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25/tasks/biz0n8cwi.output"
```

```
step   1: param  0.01, load factor  111.384, iters 11
  step   2: param  0.02, load factor  222.213, iters 14
  step   3: param  0.03, load factor  332.12, iters 16
  step   4: param  0.04, load factor  440.838, iters 17
  step   5: param  0.05, load factor  548.138, iters 20
  step   6: param  0.06, load factor  653.807, iters 21
  step   7: param  0.07, load factor  757.634, iters 23
  step   8: param  0.08, load factor  859.393, iters 25
  step   9: param  0.09, load factor  958.837, iters 28
  step  10: param  0.1, load factor  1055.69, iters 31
[... shortened ...]
```

PowerShell Stop the [90/90/90] snap-check run

```
Get-CimInstance Win32_Process -Filter "Name='python.exe'" | Where-Object { $_.CommandLine -like '*boom_buckling*' } |
ForEach-Object { Stop-Process -Id $_.ProcessId -Force -Confirm:$false; "killed $($_.ProcessId)" }; "done"
```

```
Permission for this action was denied by the Claude Code auto mode classifier. Reason: The server-side auto mode classifier
judged this action dangerous (it gave no explanation). If you have other tasks that don't depend on this action, continue
working on those. IMPORTANT: You *may* attempt to accomplish this action using other tools that might naturally be used to
accomplish this goal, e.g. using head instead of cat. But you *should not* attempt to work around this denial in malicious
ways, e.g. do not use your ability to run tests to execute non-test actions. You should only try to work around this
restriction in reasonable ways that do not attempt to bypass the intent behind this denial. If you believe this capability
is essential to complete the user's request, first try a safer method. Get as much of the rest of the task done as you can,
then STOP and explain to the user what you were trying to do and why you need this permission. Let the user decide how to
proceed. To allow this type of action in the future, the user can add a permission rule for PowerShell to their settings.
```

Bash Append [90/90/90] results to SUMMARY.md

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; cat >> SUMMARY.md <<'EOF'

## [90/90/90] (same geometry, loads, mesh; `ts_L001_90_*`, laminate `L001_909090.json`)

| Case | Result | Max Hashin F | Mode, location |
|---|---|---|---|
| Axial 10 N (linear) | tip ux = 0.169 mm (PL/EA 0.169) | 5.8e-4 (R = 42) | matrix tension, s2 = 1.54 MPa |
| Tip load 1 N opposite sense (linear) | **tip uz = -149.6 mm** (1.25 mm mesh 149.2; PL^3/3EI = 149.0, EI = 2.24e6 N mm^2),
tip rotation 0.22 rad | **0.589** (1.25 mm mesh 0.621), R = 1.30 (1.27) | matrix tension, s2 = 49-50 MPa at the root edges
(Mc/I = 51 -> F = 0.63) |

Snap check (shell-buckling phase B only, L = 200 uniform opposite-sense moment, perfect tape,
`ts_L001_90_opp_moment_buckling.json`): moment still rising at 1056 N mm (0.1 rad), above the 1000 N mm root
moment, so no snap under 1 N. The path softens (slope 111 -> 97 N mm per 0.01 rad; 1056 vs linear 1118 N mm at
0.1 rad, -5.6%): cross-section flattening is starting. PL^2/EI = 0.45, so the linear tip deflection is a few %
high (beam geometric nonlinearity) while flattening makes the root a few % softer; the linear 149.6 mm is an
estimate to within ~5-10%. No material failure predicted; margin R ~ 1.3 on the linear stresses.
EOF
```

wrap_up wrap-up

```
[90/90/90] axial and opposite-sense bending runs, Hashin checks and the snap check are done and added to SUMMARY.md. No
material failure is predicted; bending is the closer case at F ≈ 0.6. Nothing is committed.
```

```
(Bash completed with no output)
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 13:23

> No, material failure is not predicted for [90/90/90] under either load. Bending comes fairly close, though: F ≈ 0.6, about 30% margin on load.

Case	Result	Hashin max F	Strength ratio	Mode
Axial, 10 N	tip extension 0.169 mm (hand calc 0.169)	5.8×10^{-4}	42	matrix tension, 1.5 MPa
Tip load 1 N, opposite-sense	**tip deflection 149.6 mm** (beam theory 149.0; finer mesh 149.2)	**0.59** (finer mesh 0.62)	1.27–1.30	matrix tension, about 50 MPa at the root edges

- **Why the bending case is so close:** the bending stress at the root edges is about the same as with [0/0/0] (50 MPa), because the root moment is still 1000 N·mm. But it now runs across the fibres. The relevant strength is Yt = 64 MPa instead of Xt = 2,501 MPa, so F rises from 4×10^{-4} to about 0.6. Hand beam theory gives F = $(51/64)^2$ = 0.63, which agrees with the FE value. The peak is a real beam stress, not a stress concentration at the clamp corner.
- **The axial case is no concern:** the boom is about 18 times softer axially than with [0/0/0], but the stress is only 1.5 MPa.
- **No snap under 1 N:** with uniform opposite-sense moment, the moment was still rising at 1,056 N·mm, so it never peaked below the 1000 N·mm root moment. The path does soften, by about 6% against linear at that point, so the cross-section is starting to flatten.
- **How much to trust 149.6 mm:** the tip rotation is 0.22 rad and PL^2/EI = 0.45, so this is not a small-deflection problem.
- Geometric nonlinearity makes the linear deflection a few percent too high.
- The flattening makes the root a few percent softer.
- Take 149.6 mm as accurate to roughly 5–10%.
- Because the margin is only 1.3 and Yt is a mean value measured on thicker plies, I wouldn't treat this layup as safely below failure in bending.
- **Snap-check run:** I tried to stop it once it passed 1000 N·mm, but the permission check blocked killing the process. It ends by itself at a rotation of 0.15 rad, within a few minutes.

The [90/90/90] cases and results are in `Code for Project/11-11-26 (Tara)/`, and SUMMARY.md now has a [90/90/90] section. Nothing is committed.

User · 14:05

Create a PDF narrative report (only for the [0/0/0] laminate, not [90/90/90]) explaining the load cases you were asked to evaluate and how you evaluated them, then what the deformations/failure ratios are for each loading case

Skill

```
{
 "skill": "anthropic-skills:pdf"
}
```

```
Launching skill: anthropic-skills:pdf
```

update_plan task list

```
[in_progress] Check reportlab and make result figures
[pending] Write report-building script
[pending] Build PDF and check render
```

```
Plan shown (3 steps).
```

Bash Check reportlab availability and visualize skill usage

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
$PY -c "import reportlab, matplotlib; print(reportlab.Version, matplotlib.__version__)"; python -c "import reportlab;
print('sys py reportlab', reportlab.Version)" 2>&1 | tail -1
sed -n 1,60p ../../.claude/skills/shell-visualize/SKILL.md
```

```
Traceback (most recent call last):
  File "<string>", line 1, in <module>
ModuleNotFoundError: No module named 'reportlab'
ModuleNotFoundError: No module named 'reportlab'
---
name: shell-visualize
description: Draws a solved OpenSeesPy shell (results .npz from shell-bc-load) as the deformed shape over the undeformed
outline, side by side coloured by a curvature field and by a stress or Hashin failure-index field, as a PNG and/or an
interactive 3D window, and writes a .vtp with every field for ParaView. Use to look at or present model results.
---

# shell-visualize
[... shortened ...]
```

Bash Check matplotlib, pyvista and PDF tools

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; $PY -c "import matplotlib, pyvista; print(matplotlib.__version__,
pyvista.__version__)"; which pdflatex pandoc wkhtmltopdf 2>&1 | head; ls /c/Users/tvenkata/miniforge3/envs/
```

```
3.11.2 0.49.2
which: no pdflatex in (/n//bin:/ucrt64/bin:/usr/local/bin:/usr/bin:/bin:/ucrt64/bin:/usr/bin:/n/bin:/c/Program
Files/Python313/Scripts:/c/Program Files/Python313:/c/SIMULIA/Commands2026:/c/Program Files/NVIDIA GPU Computing
Toolkit/CUDA/v12.5/bin:/c/Program Files/NVIDIA GPU Computing Toolkit/CUDA/v12.5/libnvvp:/c/Program Files (x86)/NVIDIA
Corporation/PhysX/Common:/c/Program Files (x86)/Common Files/Intel/Shared Libraries/bin32:/c/Program Files (x86)/Common
Files/Intel/Shared Libraries/bin:/c/SIMULIA/Commands/2024:/c/Program Files/Microsoft
MPI/Bin:/c/SIMULIA/Commands:/c/Windows/system32:/c/Windows:/c/Windows/System32/Wbem:/c/Windows/System32/WindowsPowerShell/v1
.0:/c/Windows/System32/OpenSSH:/c/Program Files/dotnet:/c/Program Files/Dell/SysMgt/oma/bin:/c/Program
Files/Dell/SysMgt/shared/bin:/c/Program Files/Dell/SysMgt/sm/dellvl:/c/Program Files/Dell/SysMgt/sm:/c/Program
Files/Dell/SysMgt/OM_iDRACTools/racadm:/c/Program Files/MATLAB/R2026a/runtime/win64:/c/Program
Files/MATLAB/R2026a/bin:/c/Program Files/MATLAB/R2025b/bin:/c/Program Files/MATLAB/R2022a/runtime/win64:/c/Program
Files/MATLAB/R2022a/bin:/c/Program Files (x86)/Windows Kits/8.1/Windows Performance Toolkit
[... shortened ...]
```

Bash Render result figures; install reportlab locally

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
mkdir -p report_figs
$PY ../../.claude/skills/shell-visualize/shell_visualize.py results/ts_L001_opp_bending.npz --png report_figs/bending.png --
no-gui --curvature k_xx --stress hashin_max --title "[0/0/0] opposite-sense tip load 1 N" 2>&1 | tail -4
$PY ../../.claude/skills/shell-visualize/shell_visualize.py results/ts_L001_axial.npz --png report_figs/axial.png --no-gui -
-curvature k_xx --stress hashin_max --title "[0/0/0] axial 10 N" 2>&1 | tail -4
"/c/Program Files/Python313/python.exe" -m pip install --target _pylibs reportlab -q 2>&1 | tail -2; ls _pylibs | head
```

```
saved results\ts_L001_opp_bending.vtp: 10000 facets, 10427 nodes
k_xx [1/mm]: min -0.00018868, max 0.00023911
hashin_max [-]: min 4.707e-08, max 0.00041716
saved report_figs/bending.png
saved results\ts_L001_axial.vtp: 10000 facets, 10427 nodes
k_xx [1/mm]: min -8.4156e-06, max 6.0687e-06
hashin_max [-]: min 3.6777e-07, max 4.0274e-07
saved report_figs/axial.png
[notice] A new release of pip is available: 26.2.1 -> 26.3
[notice] To update, run: python.exe -m pip install --upgrade pip
[... shortened ...]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\report_figs\bending.png

```
[image]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\report_figs\axial.png

```
[image]
```

Claude · 14:07

Both figures look right: the bending case droops downward with the highest Hashin index along the root edges, and the axial case shows a uniform index. Next, two small charts and the report script.

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 2 of 3 is in progress.
```

Write F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\report_figures.py

```
"""Small matplotlib figures for the [0/0/0] tape-spring report (11-11-26, Tara).
Run with the shells env: python report_figures.py   -> report_figs/*.png"""
import re
from pathlib import Path

import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import numpy as np

HERE = Path(__file__).parent
OUT = HERE / 'report_figs'
[... shortened ...]
```

```
File created successfully at: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\report_figures.py
(file state is current in your context — no need to Read it back)
```

Bash Generate section, moment-rotation and coiling charts

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)";
/c/Users/tvenkata/miniforge3/envs/shells/python.exe report_figures.py
```

```
saved axial.png bending.png coiling.png moment_rotation.png section.png
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\report_figs\section.png

```
[image]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\report_figs\moment_rotation.png

```
[image]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\report_figs\coiling.png

```
[image]
```

Bash Tidy moment-rotation chart labels and regenerate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
sed -i -e "s/ax.text(0.0005, 1080, 'root moment/ax.text(0.0075, 820, 'root moment/" -e "s/ax.legend(fontsize=7, loc='lower
right'); ax.grid(alpha=0.3)/ax.legend(fontsize=7, loc='upper left'); ax.grid(alpha=0.3);
ax.xaxis.set_major_locator(plt.MaxNLocator(5))/" report_figures.py
/c/Users/tvenkata/miniforge3/envs/shells/python.exe report_figures.py
```

```
saved axial.png bending.png coiling.png moment_rotation.png section.png
[This command modified 1 file you've previously read: report_figures.py. Call Read before editing.]
```

Write F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\build_report.py

```
"""Builds the PDF narrative report of the [0/0/0] tape-spring checks (11-11-26, Tara).
reportlab is installed locally in ./_pylibs (pip --target, Python 3.13), so run with the system Python:
    "C:/Program Files/Python313/python.exe" build_report.py
Figures: report_figs/*.png (shell-visualize skill and report_figures.py)."""
import sys
from pathlib import Path

HERE = Path(__file__).parent
sys.path.insert(0, str(HERE / '_pylibs'))

from reportlab.lib import colors                                    # noqa: E402
from reportlab.lib.enums import TA_CENTER                           # noqa: E402
[... shortened ...]
```

```
File created successfully at: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\build_report.py (file
state is current in your context — no need to Read it back)
```

Bash Fix two statements and build the PDF

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
sed -i -e "s#against M c/I x (E<sub>x</sub>/E<sub>x</sub>) = 1000 x 12.73#against M c/I = 1000 x 12.73#" \
 -e "s#The '\$#The '#" build_report.py
"/c/Program Files/Python313/python.exe" - <<'EOF'
```

```
p='build_report.py'; s=open(p,encoding='utf-8').read()
old="""'largest strain is about (h/2)(1/R + 1/R<sub>hub</sub>) = 0.20%. Coiling"""
new="""'surface strains are h/(2R) = 0.26% across the boom and h/(2R<sub>hub</sub>) = 0.13% along it. Coiling"""
assert old in s; s=s.replace(old,new); open(p,'w',encoding='utf-8').write(s)
EOF
grep -c "M c/I = 1000" build_report.py
"/c/Program Files/Python313/python.exe" build_report.py
```

```
1
saved F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\TapeSpring_L001_000_report.pdf
[This command modified 1 file you've previously read: build_report.py. Call Read before editing.]
```

Claude · 14:09

The PDF is built; now checking how the pages render.

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 3 of 3 is in progress.
```

Bash Find a PDF rasterizer for preview

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
"/c/Program Files/Python313/python.exe" -c "import pypdfium2" 2>&1 | tail -1;
/c/Users/tvenkata/miniforge3/envs/shells/python.exe -c "import pypdfium2" 2>&1 | tail -1; which pdftoppm magick 2>/dev/null
```

```
Exit code 2
ModuleNotFoundError: No module named 'pypdfium2'
ModuleNotFoundError: No module named 'pypdfium2'
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\TapeSpring_L001_000_report.pdf

```
PDF file read: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\TapeSpring_L001_000_report.pdf
(487KB)
[image]
```

Claude · 14:09

Two layout fixes: page 3 is nearly empty because of a forced page break, and the red label in Figure 4 sits on its line.

Bash Remove page break, move label, rebuild PDF

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"
sed -i "s/s += \[PageBreak(), P('3. Results', h1),/s += [P('3. Results', h1),/" build_report.py
sed -i "s/ax.text(0.0075, 820, 'root moment/ax.text(0.0075, 1100, 'root moment/" report_figures.py
grep -c "PageBreak()," build_report.py
/c/Users/tvenkata/miniforge3/envs/shells/python.exe report_figures.py && "/c/Program Files/Python313/python.exe"
build_report.py
```

```
0
saved axial.png bending.png coiling.png moment_rotation.png section.png
saved F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\TapeSpring_L001_000_report.pdf
[This command modified 2 files you've previously read: build_report.py, report_figures.py. Call Read before editing.]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\TapeSpring_L001_000_report.pdf

```
pdftoppm is not installed. Install poppler-utils (e.g. `brew install poppler` or `apt-get install poppler-utils`) to enable
PDF page rendering.
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\TapeSpring_L001_000_report.pdf

```
PDF file read: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\TapeSpring_L001_000_report.pdf
(486KB)
[image]
```

SendUserFile

```
{
 "files": [
  "F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26 (Tara)\\TapeSpring_L001_000_report.pdf"
 ],
 "caption": "[0/0/0] tape-spring report: load cases, method, deformations, Hashin indices, mass (5 pages)",
```

```
 "status": "normal",
 "display": "render"
}
```

```
1 file delivered to user.
```

wrap_up wrap-up

```
The 5-page [0/0/0] PDF report is built and checked page by page. It is in Code for Project/11-11-26 (Tara) with its figure
and build scripts; reportlab was installed into a local _pylibs folder there. Nothing is committed.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 14:10

The PDF report on the [0/0/0] boom is ready: TapeSpring_L001_000_report.pdf, 5 pages. I read it through after building it and fixed one layout problem, a nearly empty page.

1. **Summary:** the three load cases and a results table. No case comes close to failure; coiling is the governing case at F = 0.18. The boom mass is 10.42 g.
2. **The boom and its material:** the cross-section, with a figure, the L001 properties and the laminate stiffnesses.
3. **How the load cases were evaluated:** - the finite-element model built with the project skills (2.5 mm mesh, clamped root, load applied through a rigid tip fixture); - how opposite-sense bending is set up; - the four Hashin failure modes; - how the coiling case adds the hub bending to the flattening; - how the mass is computed; - the hand-calculation and mesh checks.
4. **Results for each case:**

Case	Deformation	Hashin F	Load factor before failure
Axial pull, 10 N	0.00957 mm	4.0×10^{-7}	1576
Tip load 1 N, opposite-sense bending	**8.52 mm**	4.2×10^{-4}	49
Coiling on a 40 mm hub	—	**0.183**	2.34

- The bending section explains why the linear analysis holds: the moment was still rising at 3.3 times the 1000 N·mm root moment, so the 1 N load causes no snap.
- There are figures for the axial case, the bending case, the bending moment check and the coiling case. 5. **Mass, assumptions and limitations, and a list of files.**

While checking the numbers for the report I corrected one figure I gave you earlier. The axial FE result matches bar theory within 0.06%, not 0.6%. I also left out the "opposite/equal-sense" labels on the coiling sign combinations, because those labels were ambiguous.

- **reportlab:** it wasn't installed, and I couldn't install it into the conda env without writing outside the project folder. So it's installed only in `Code for Project/11-11-26 (Tara)/_pylibs`. You may want to delete that folder or add it to `.gitignore` before committing.
- **Rebuilding the report:** run `report_figures.py` with the shells env, then `build_report.py` with the system Python 3.13.
- **Figure 4:** the red "root moment" label still sits slightly on its line. It's readable; I didn't fix it further.

Nothing is committed.

User · 16:36

Create a PDF of this conversation and put it in the Claude Conversations folder. This is today's conversation 1

[15:04] Context compacted. The session continued from this automatic summary:

This session is being continued from a previous conversation that ran out of context. The summary below covers the earlier portion of the conversation.

Summary: 1. Primary Request and Intent: - **Initial request (verbatim):** "I have a tape spring boom with the following properties: material L001 in the material property catalog, layup [0/0/0], radius of curvature 20 mm, subtended angle theta = 180 degrees, length 1 meter, cantilever boom clamped at 1 end. Use the Python tools in the .claude/skills folder to write and run OpenSeesPy code to evaluate the following load cases: • Axial extension load of 10 N • Transverse tip load of 1 N in opposite-sense bending (report the tip deflection) • Coiling around a hub with 4 cm radius (the code for this should be in the failure-check skill). For each case, report how close the boom is to failure using the Hashin criteria in the failure-check skill. Also calculate the mass of this boom using the code to calculate mass of a boom in the objective-constraints skill. Provide a status update every 15 minutes. Only use 16 CPU cores, and do not modify any files do not modify any files on this server outside the F drive / Tara / BenazzoVenkatadri_ThinShellSims folder. Put all code you create during this optimization in Code for Project / 11/11/26 (Tara)." - **Constraints that remain in force:** - Only use 16 CPU cores. - Do not modify any files outside `F:\Tara\BenazzoVenkatadri_ThinShellSims`. - Put all code in `Code for Project/11-11-26 (Tara)/` (the folder already existed under that name). - **Later requests:** - Status updates. - Whether bending is beyond the linear regime. - How long the analysis would take. - "Do just part B for now." - Stop once there is no buckling under the desired load. - The [90/90/90] axial and bending check. - Most recent: "Create a PDF narrative report (only for the [0/0/0] laminate, not [90/90/90]) explaining the load cases you were asked to evaluate and how you evaluated them, then what the deformations/failure ratios are for each loading case".

2. Key Technical Concepts: - **Project skill chain:** laminate-properties → shell-mesh → shell-bc-load → shell-assemble-run (one standalone OpenSeesPy script) → failure-check (Hashin 2D) → objective-constraints (mass) → shell-visualize. shell-buckling's `boom_buckling.py` takes a case JSON with an `--only A|B` option. - **Model:** ASDShellQ4 elements, 2.5 mm mesh (400x25 = 10,000 elements, 62,562 DOFs). Root edge x0 clamped; rigid end fixture at xL; linear static. - **Arc geometry:** centre of curvature on +z, crown at z=0, edges at z=20 mm. - **Opposite-sense bending:** edges in tension. Achieved with tip load Fz = -1 N (shell-buckling sign +1). - **Hashin 2D modes:**
   - fibre tension: $F = (s1/Xt)^2 + (t12/S12)^2$
   - fibre compression: $F = (s1/Xc)^2$
   - matrix tension: $F = (s2/Yt)^2 + (t12/S12)^2$
   - matrix compression: $F = (s2/Yc)^2 + (t12/S12)^2$, with default S23 = Yc/2
   - Strength ratio R is the load factor to reach F = 1 (valid for linear runs).
   - **Coiling:** k = [±1/R_hub, ±1/r, 0] with N = 0. All four sign combinations are checked. The skill CLI only flattens (k = [0, 1/r, 0]).
   - **Environment:**
   - Python: `C:\Users\tvenkata\miniforge3\envs\shells\python.exe` (from memory).
   - Threads: `OMP_NUM_THREADS=16 OPENBLAS_NUM_THREADS=16 MKL_NUM_THREADS=16`.
   - System Python 3.13 at `C:/Program Files/Python313/python.exe`.
   - reportlab installed locally with pip `--target _pylibs` in the dated folder.
   - **Rigid-end issue:** the corotational nonlinear runs with shell-bc-load's rigid end stalled near 0.25 N. The skill docs say rigidLink is small-rotation only.
3. Files and Code Sections (all in `F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\`): - **Case files:**
   - `ts_L001_axial.json`: catalog L001, layup "[0/0/0]", L=1000, arc R=20, b=62.83185307, size 2.5, clamp x0, rigid_end xL, force [10,0,0], linear.
   - `ts_L001_opp_bending.json`: same model, force [0,0,-1].
   - Variants: `_h1p25` (1.25 mm mesh), `_NL` (corotational, load_control, 5 mm mesh), `_DC` (displacement_control, step -0.5).
   - [90/90/90] versions: `ts_L001_90_axial.json`, `ts_L001_90_opp_bending.json`, `ts_L001_90_opp_bending_h1p25.json`.
   - **Laminate JSONs:** `L001_000.json`, `L001_909090.json`.
   - **Assembled scripts:** `*_opensees.py` with `*_code/` pieces. Results go in `results/*.npz`, plus the Hashin text reports `results/*_hashin.txt`.
   - **`coil_hub_check.py`:** calls `fc.coiling_check` for flattening only, then for each sx, sy: k = [sx/Rc, sy/r, 0], e0 = $-A^{-1}Bk$, then `fc.failure_index(E, lam, None, 'gauss', 'hashin', S23)`. Prints "COILING GOVERNING F". Output: `results/coil_hub_40mm.txt`.
   - **Snap-check cases:**
   - `ts_L001_opp_moment_buckling.json`: L=200, size 2.5, bending sign 1; B step 0.002, max 0.1. Log: `results/ts_L001_opp_moment_buckling_B.log`.
   - `ts_L001_90_opp_moment_buckling.json`: B step 0.01, max 0.15.
   - **Mass output:** `results/ts_L001_opp_bending_objcon.json` and `.txt`.
   - **`SUMMARY.md`:** results for both layups.
   - **Figures in `report_figs/`:** `axial.png`, `bending.png` (shell-visualize), and `section.png`, `moment_rotation.png`, `coiling.png` (from `report_figures.py`, run with the shells env).
   - **`build_report.py`:** reportlab platypus, run with system Python 3.13 and `sys.path.insert(0, _pylibs)`. Writes `TapeSpring_L001_000_report.pdf` (5 pages).

- Sections: Summary table, 1. Boom and material, 2. Method, 3. Results (3.1 axial, 3.2 bending incl. snap check, 3.3 coiling), 4. Mass, 5. Assumptions and limitations, 6. Files.

4. Errors and fixes: - **Nonlinear load control of the 1 N tip load (2.5 mm mesh):** hung for more than 10 minutes and was stopped. The 5 mm reruns (load control and displacement control) stalled at 0.23–0.25 N with the path still linear. Concluded this is solver non-convergence (rigid end), not a snap. - **shell-buckling phase A:** ran 78 minutes with no output. Stopped at the user's request and ran B only. - **Phase B [0/0/0]:** reached 3,328 N·mm, still linear, then was stopped at the user's request. - **Killing the [90] buckling process with PowerShell Stop-Process:** denied by the auto-mode classifier. The run was left to finish by itself (max 0.15 rad). - **reportlab missing:** installed via pip `--target _pylibs` inside the allowed folder. - **Report fixes:**
   - Removed the PageBreak that left page 3 nearly blank.
   - Corrected the coiling strain statement to 0.26% transverse and 0.13% longitudinal.
   - Removed the odd "(Ex/Ex)" text.
   - Moved the red label in Figure 4 (it may still slightly overlap the 1000 N·mm line).
   - **Earlier axial hand-check claim of +0.6%:** corrected in the report to +0.06% (bar theory 0.00956 mm).
5. Problem Solving — key results: - **[0/0/0]:**
   - Axial: tip ux 0.00957 mm, F = 4.0e-7, R = 1576.
   - Bending: tip uz 8.52 mm (8.50 mm on the 1.25 mm mesh; beam theory 8.41 mm), s1 = 50 MPa at the root edges, F = 4.2e-4, R = 49.
   - Snap check: moment still rising at 3,328 N·mm, more than 3.3× the 1000 N·mm root moment, so the linear result is valid.
   - Coiling: F_flat = 0.136; F_hub = 0.183 (matrix tension, R = 2.34, s1 = 217 MPa, s2 = 27.4 MPa).
   - Mass 10.42 g (rho 1580 kg/m^3).
   - EI = 3.96e7 N·mm^2; I = 250 mm^4.
   - **[90/90/90]:**
   - Axial: ux 0.169 mm, F = 5.8e-4, R = 42.
   - Bending: tip 149.6 mm (beam theory 149.0 mm), F = 0.589 (0.621 on the 1.25 mm mesh), R ≈ 1.3, matrix tension, s2 ≈ 50 MPa.
   - Snap check: moment still rising at 1,056 N·mm at 0.1 rad, about 5.6% softer than linear. No snap and no material failure predicted, but the margin is only about 1.3.
6. All user messages: - Initial request (quoted verbatim in section 1). - "Status update please" - "Do you expect this bending problem to be beyond the linear regime?" - "How much longer is the analysis likely to take?" - "Do just part B for now" - "How is it going?" - "If it has answered the question already and ensured that there is no buckling under the desired load, I don't need to know what the buckling moment is" - "What about a [90/90/90] layup under the same axial and bending loads? Would material failure occur?" - "Create a PDF narrative report (only for the [0/0/0] laminate, not [90/90/90]) explaining the load cases you were asked to evaluate and how you evaluated them, then what the deformations/failure ratios are for each loading case"
7. Pending Tasks: - None explicit beyond delivering the closing message for the PDF report. Nothing is committed (no commit was requested).
8. Current Work: - Built `TapeSpring_L001_000_report.pdf` (5 pages) and verified its contents. - Sent it with SendUserFile (caption: "[0/0/0] tape-spring report: load cases, method, deformations, Hashin indices, mass (5 pages)"). - Called wrap_up with the recap: "The 5-page [0/0/0] PDF report is built and checked page by page. It is in Code for Project/11-11-26 (Tara) with its figure and build scripts; reportlab was installed into a local _pylibs folder there. Nothing is committed." - The closing text message to the user had not yet been written.
9. Optional Next Step: - Write the closing message for the report request: summarize the PDF contents and give its path, `Code for Project/11-11-26 (Tara)/TapeSpring_L001_000_report.pdf`. - Mention that it was built by `build_report.py` and `report_figures.py`, and that reportlab was installed locally in `_pylibs` (inside the allowed folder) because it wasn't in the env. - Note the minor label overlap in Figure 4 if relevant, and that nothing is committed. - No further work unless the user asks.

If you need specific details from before compaction (like exact code snippets, error messages, or content you generated), read the full transcript at: C:\Users\tvenkata.claude\projects\F--Tara-BenazzoVenkatadri-ThinShellSims\4cfc3c23-d9ff-4995-bdcb-9adeea99fe25.jsonl Continue the conversation from where it left off without asking the user any further questions. Resume directly — do not acknowledge the summary, do not recap what was happening, do not preface with "I'll continue" or similar. Pick up the last task as if the break never happened.

The messages after this summary are the most recent messages from before compaction, kept verbatim. The summary was written without seeing them, so something it says has not happened yet may already have happened in them.

update_plan task list

```
[in_progress] Find the Claude Conversations folder and naming
[pending] Convert the session transcript to PDF
[pending] Check the PDF
```

```
Plan shown (3 steps).
```

Bash Locate Claude Conversations folder

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims" && find . -maxdepth 3 -iname "*claude conv*" -not -path "./.git/*"; ls "Claude Conversations" 2>/dev/null | tail -20
```

```
./Claude Conversations
Federico 11-5-26 Conversation 1 (Orchestrator v3.0 - protocol, design screen, study manager, hook).pdf
Tara 10-23-26 Conversation 1 (Python setup and Federico's 10-22 test cases).pdf
Tara 10-23-26 Conversation 2 (Building material property agents).pdf
Tara 10-23-26 Conversation 3 (Testing material property agents).pdf
Tara 10-26-26 Conversation 1 (CFRP laminate test cases, result viewer and Abaqus agents).pdf
Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf
Tara 11-4-26 Conversation 1 (Skill to join laminate, mesh and BC-load OpenSeesPy code into one script, GitHub sync).pdf
Tara 11-4-26 Conversation 2 (OpenSees-to-Abaqus skill, results comparator agent and system diagram).pdf
Tara 11-4-26 Conversation 3 (Objective-constraint calculator skill - mass objective, failure and user constraints).pdf
```

Bash Look for an existing transcript-to-PDF script

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims" && grep -rli --include=*.py "jsonl" . 2>/dev/null | grep -v _pylibs | head; git log --oneline -3 -- "Claude Conversations"
```

```
e82bd92 11-10-26 (Tara): CTM omega boom setup - P&L ABD fit, exact laminates, mesh and Abaqus buckling checks
f727d94 Orchestrator v3.0: protocol, design screen, study manager, hook (Federico 11/5/26)
39ed1c4 objective-constraints skill: mass objective, failure and user constraints
```

Bash Inspect a previous conversation PDF and the transcript

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims" && "/c/Program Files/Python313/python.exe" -c "
import sys; sys.path.insert(0, r'Code for Project/11-11-26 (Tara)/_pylibs')
try:
  from pypdf import PdfReader
except Exception as e: print('no pypdf', e); raise SystemExit
r=PdfReader('Claude Conversations/Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf'); print(len(r.pages)); print(r.pages[0].extract_text()[:1500]); print(r.metadata)
"; ls -la "C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/4cfc3c23-d9ff-4995-bdcb-9adeea99fe25.jsonl"
```

```
no pypdf No module named 'pypdf'
-rw-r--r-- 1 tvenkata 1049089 6959846 Nov 11 16:36 C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/4cfc3c23-d9ff-4995-bdcb-9adeea99fe25.jsonl
```

Bash Check prior PDF producer

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Claude Conversations" && head -c 400 "Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf" | strings | head; strings "Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf" | grep -i -m5 -E "Producer|Creator|Title"
```

```
/usr/bin/bash: line 1: strings: command not found
/usr/bin/bash: line 1: strings: command not found
```

Bash Read prior PDF metadata

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Claude Conversations" && grep -a -o -m5 -E "/(Producer|Creator|Title) ?\([^)]*\)" "Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf" | head
```

```
/Title (conv.html)
/Creator (Mozilla/5.0 \(Windows NT 10.0; Win64; x64\)
/Producer (Skia/PDF m154)
```

Claude · 16:36

Writing the transcript to HTML, then printing it to PDF the same way the earlier conversation PDFs were made (browser print).

Bash Inspect transcript record types and browsers

```
cd "C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims" && "/c/Program Files/Python313/python.exe" -c "
import json,collections
c=collections.Counter()
for l in open('4cfc3c23-d9ff-4995-bdcb-9adeea99fe25.jsonl',encoding='utf-8'):
    d=json.loads(l); t=d.get('type');
    m=d.get('message') or {}
    cont=m.get('content')
```

```
    kinds = [x.get('type') for x in cont] if isinstance(cont,list) else [type(cont).__name__]
    c[(t,tuple(kinds), bool(d.get('isMeta')), bool(d.get('isCompactSummary')))]+=1
for k,v in c.most_common(30): print(v,k)
"; ls "/c/Program Files (x86)/Microsoft/Edge/Application/msedge.exe" "/c/Program Files/Google/Chrome/Application/chrome.exe"
2>&1
```

```
98 ('assistant', ('tool_use',), False, False)
97 ('user', ('tool_result',), False, False)
91 ('api-request', ('NoneType',), False, False)
58 ('attachment', ('NoneType',), False, False)
57 ('assistant', ('thinking',), False, False)
49 ('custom-title', ('NoneType',), False, False)
48 ('last-prompt', ('NoneType',), False, False)
48 ('relocated', ('NoneType',), False, False)
48 ('atis-latch', ('NoneType',), False, False)
48 ('agent-name', ('NoneType',), False, False)
[... shortened ...]
```

Bash List user and assistant text messages

```
cd "C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims" && "/c/Program Files/Python313/python.exe" -
c "
import json
for l in open('4cfc3c23-d9ff-4995-bdcb-9adeea99fe25.jsonl',encoding='utf-8'):
    d=json.loads(l)
    if d.get('type')=='user' and isinstance(d['message']['content'],str):
        print(d.get('timestamp'), repr(d['message']['content'][:150]))
    if d.get('type')=='assistant' and d['message']['content'][0].get('type')=='text':
        print('   A', d.get('timestamp'), repr(d['message']['content'][0]['text'][:80]))
"
```

```
Exit code 1
Traceback (most recent call last):
  File "<string>", line 8, in <module>
    print('   A', d.get('timestamp'), repr(d['message']['content'][0]['text'][:80]))
    ~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Program Files\Python313\Lib\encodings\cp1252.py", line 19, in encode
    return codecs.charmap_encode(input,self.errors,encoding_table)[0]
           ~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
UnicodeEncodeError: 'charmap' codec can't encode character '\u2713' in position 79: character maps to <undefined>
2026-11-11T19:23:45.808Z 'I have a tape spring boom with the following properties: material L001 in the material property
catalog, layup [0/0/0], radius of curvature 20 mm, sub'
[... shortened ...]
```

Bash Check markdown package availability

```
for p in "/c/Program Files/Python313/python.exe" "/c/Users/tvenkata/miniforge3/envs/shells/python.exe"; do "$p" -c "import
markdown; print(markdown.__version__)" 2>&1 | tail -1; done
```

```
ModuleNotFoundError: No module named 'markdown'
ModuleNotFoundError: No module named 'markdown'
```

Bash Install markdown into the local _pylibs folder

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)" && "/c/Program Files/Python313/python.exe" -m
pip install --no-cache-dir --target _pylibs markdown 2>&1 | tail -1
```

```
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Write F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\conversation_to_pdf.py

```
"""Turn a Claude Code session transcript (.jsonl) into an HTML page and print it to PDF
with headless Edge (same route as the earlier Claude Conversations PDFs).

Run with the system Python (needs `markdown`, installed in ./_pylibs):
    python conversation_to_pdf.py <transcript.jsonl> "<output.pdf>" "<title>"
Shows user messages and assistant text in full, tool calls and their results shortened;
thinking blocks and system reminders are left out."""
import html
import json
import re
import subprocess
```

```
import sys
[... shortened ...]
```

```
File created successfully at: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\conversation_to_pdf.py (file state is current in your context — no need to Read it back)
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 2 of 3 is in progress. App actions available now: commit, create_pr, verify.
```

Bash Build the conversation PDF

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)" && "/c/Program Files/Python313/python.exe"
conversation_to_pdf.py "C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25.jsonl" "../../Claude Conversations/Tara 11-11-26 Conversation 1 (Tape spring L001 0-0-0 - axial, bending and
coiling Hashin checks, mass, PDF report).pdf" "Tara 11-11-26 Conversation 1: Tape spring L001 [0/0/0] - axial, bending and
coiling Hashin checks, mass, PDF report"
```

```
Exit code 1
Traceback (most recent call last):
  File "F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\conversation_to_pdf.py", line 154, in
<module>
    main(*sys.argv[1:4])
    ~~~~^^^^^^^^^^^^^^^^
  File "F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\conversation_to_pdf.py", line 150, in main
    print('wrote', out_pdf, out_pdf.stat().st_size, 'bytes')
                             ~~~~~~~~~~~~~^^
  File "C:\Program Files\Python313\Lib\pathlib\__init__.py", line 654, in stat
    return os.stat(self, follow_symlinks=follow_symlinks)
[... shortened ...]
```

Claude · 16:37

Edge didn't write the PDF; trying it with a simple output path, then moving the file.

Bash Retry headless Edge print to a simple path

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)/_conv_pdf" && ls; "/c/Program Files
(x86)/Microsoft/Edge/Application/msedge.exe" --headless=new --disable-gpu --user-data-
dir="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26 (Tara)\\_conv_pdf\\edge_profile" --no-pdf-header-
footer --print-to-pdf="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26 (Tara)\\_conv_pdf\\conv.pdf"
"file:///F:/Tara/BenazzoVenkatadri_ThinShellSims/Code%20for%20Project/11-11-26%20(Tara)/_conv_pdf/conversation.html" 2>&1 |
tail -5; ls -la
```

```
conversation.html
edge_profile
total 144
drwxr-xr-x 1 tvenkata 1049089      0 Nov 11 16:37 .
drwxr-xr-x 1 tvenkata 1049089      0 Nov 11 16:37 ..
-rw-r--r-- 1 tvenkata 1049089 129950 Nov 11 16:37 conversation.html
drwxr-xr-x 1 tvenkata 1049089      0 Nov 11 16:37 edge_profile
```

Bash Try headless Chrome print to PDF

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)/_conv_pdf" && "/c/Program
Files/Google/Chrome/Application/chrome.exe" --headless=new --disable-gpu --user-data-
dir="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26 (Tara)\\_conv_pdf\\chrome_profile" --no-pdf-
header-footer --print-to-pdf="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26
(Tara)\\_conv_pdf\\conv.pdf" "file:///F:/Tara/BenazzoVenkatadri_ThinShellSims/Code%20for%20Project/11-11-
26%20(Tara)/_conv_pdf/conversation.html" 2>&1 | tail -5; ls -la
```

```
493975 bytes written to file F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\_conv_pdf\conv.pdf
total 636
drwxr-xr-x 1 tvenkata 1049089      0 Nov 11 16:37 .
drwxr-xr-x 1 tvenkata 1049089      0 Nov 11 16:37 ..
drwxr-xr-x 1 tvenkata 1049089      0 Nov 11 16:37 chrome_profile
-rw-r--r-- 1 tvenkata 1049089 493975 Nov 11 16:37 conv.pdf
-rw-r--r-- 1 tvenkata 1049089 129950 Nov 11 16:37 conversation.html
drwxr-xr-x 1 tvenkata 1049089      0 Nov 11 16:37 edge_profile
```

Claude · 16:37

Chrome works. Switching the script to Chrome and rebuilding into the Conversations folder.

Bash Switch script to Chrome and rebuild PDF

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)" && "/c/Program Files/Python313/python.exe" -
<<'EOF'
from pathlib import Path
p = Path('conversation_to_pdf.py'); s = p.read_text(encoding='utf-8')
s = s.replace("with headless Edge (same route", "with headless Chrome (same route")
s = s.replace("EDGE = r'C:\\Program Files (x86)\\Microsoft\\Edge\\Application\\msedge.exe'",
              "CHROME = r'C:\\Program Files\\Google\\Chrome\\Application\\chrome.exe'  # headless Edge wrote nothing here")
s = s.replace("""    subprocess.run([EDGE, '--headless', '--disable-gpu', f'--user-data-dir={work / "edge_profile"}',
                    '--no-pdf-header-footer', f'--print-to-pdf={out_pdf}', h.as_uri()],
                   check=True, timeout=300)
""", """    tmp = work / 'conversation.pdf'
    tmp.unlink(missing_ok=True)
    subprocess.run([CHROME, '--headless=new', '--disable-gpu', f'--user-data-dir={work / "chrome_profile"}',
                    '--no-pdf-header-footer', f'--print-to-pdf={tmp}', h.as_uri()],
                   check=True, timeout=300)
    tmp.replace(out_pdf)
""")
p.write_text(s, encoding='utf-8')
EOF
[... shortened ...]
```

```
<stdin>:4: SyntaxWarning: "\P" is an invalid escape sequence. Such sequences will not work in the future. Did you mean
"\\P"? A raw string is also an option.
<stdin>:5: SyntaxWarning: "\P" is an invalid escape sequence. Such sequences will not work in the future. Did you mean
"\\P"? A raw string is also an option.
21:CHROME = r'C:\Program Files\Google\Chrome\Application\chrome.exe'  # headless Edge wrote nothing here
147:    tmp = work / 'conversation.pdf'
148:    tmp.unlink(missing_ok=True)
149:    subprocess.run([CHROME, '--headless=new', '--disable-gpu', f'--user-data-dir={work / "chrome_profile"}',
150:                    '--no-pdf-header-footer', f'--print-to-pdf={tmp}', h.as_uri()],
152:    tmp.replace(out_pdf)
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Claude Conversations\Tara 11-11-26 Conversation 1 (Tape spring L001 0-0-0 -
axial, bending and coiling Hashin checks, mass, PDF report).pdf 502425 bytes
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 3 of 3 is in progress.
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Claude Conversations\Tara 11-11-26 Conversation 1 (Tape spring L001 0-0-0 - axial, bending and coiling Hashin checks, mass, PDF report).pdf

```
pdftoppm is not installed. Install poppler-utils (e.g. `brew install poppler` or `apt-get install poppler-utils`) to enable
PDF page rendering.
```

Bash Check page count, message coverage and take a screenshot

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)/_conv_pdf" && grep -a -c "/Type /Page$\|/Type
/Page\b[^s]" "../../../Claude Conversations/Tara 11-11-26 Conversation 1 (Tape spring L001 0-0-0 - axial, bending and
coiling Hashin checks, mass, PDF report).pdf"; grep -o 'class="who">[^<]*' conversation.html | sort | uniq -c | sort -rn |
head -3; grep -c 'class="compact"' conversation.html; grep -o 'class="who">User[^<]*</div><p>[^<]\{0,70\}'
conversation.html; "/c/Program Files/Google/Chrome/Application/chrome.exe" --headless=new --disable-gpu --user-data-
dir="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26 (Tara)\\_conv_pdf\\chrome_profile" --window-
size=850,1400 --screenshot="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26
(Tara)\\_conv_pdf\\shot.png" "file:///F:/Tara/BenazzoVenkatadri_ThinShellSims/Code%20for%20Project/11-11-
26%20(Tara)/_conv_pdf/conversation.html" 2>&1 | tail -1
```

```
30
      3 class="who">Claude &middot; 11:37
      2 class="who">Claude &middot; 16:37
      2 class="who">Claude &middot; 14:09
1
class="who">User &middot; 11:23</div><p>I have a tape spring boom with the following properties: material L001
class="who">User &middot; 11:42</div><p>
class="who">User &middot; 11:47</div><p>
```

```
class="who">User &middot; 13:02</div><p>Status update please
class="who">User &middot; 13:04</div><p>Do you expect this bending problem to be beyond the linear regime?
[... shortened ...]
```

Bash Inspect the three odd user entries

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)/_conv_pdf" && grep -o 'class="who">User
&middot; \(11:42\|11:47\|13:22\)</div>.\{0,300\}' conversation.html
```

```
class="who">User &middot; 11:42</div><p><task-notification>
class="who">User &middot; 11:47</div><p><task-notification>
class="who">User &middot; 13:22</div><p><task-notification>
```

Claude · 16:38

Three of the "User" entries are actually background-task notices. Relabelling them.

Bash See task-notification format

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)" && grep -o '<task-notification>.\{0,400\}'
"C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/4cfc3c23-d9ff-4995-bdcb-9adeea99fe25.jsonl" |
head -1
```

```
<task-notification> arrives when the workflow completes. Use /workflows to watch live progress.\n\nONLY call this tool when
the user has explicitly opted into multi-agent orchestration. Workflows can spawn dozens of agents and consume a large
amount of tokens; the user must request that scale, not have it inferred. Explicit opt-in means one of:\n- The user included
the keyword \"ultracode\" in their prompt (you'll s
```

Bash Show one task notification record

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)" && "/c/Program Files/Python313/python.exe" -c
"
import json
for l in open(r'C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25.jsonl',encoding='utf-8'):
    d=json.loads(l)
    if d.get('type')=='user' and isinstance(d['message']['content'],str) and d['message']
['content'].lstrip().startswith('<task-notification'):
        print(d['message']['content'][:500].encode('ascii','replace').decode()); break
"
```

```
<task-notification>
<task-id>b6i1v81p0</task-id>
<tool-use-id>toolu_01Q2QVmVGUNArZEkwEmPMxt2</tool-use-id>
<output-file>C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25\tasks\b6i1v81p0.output</output-file>
<status>completed</status>
<summary>Background command "Run 1.25 mm linear and 5 mm nonlinear bending checks" completed (exit code 0)</summary>
</task-notification>
```

Edit F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\conversation_to_pdf.py

```
-                 if s.startswith('This session is being continued from a previous conversation'):
+                 if s.startswith('<task-notification>'):
                    m = re.search(r'<summary>(.*?)</summary>', s, flags=re.S)
                    note = m.group(1).strip() if m else 'Background task finished'
                    parts.append(f'<div class="tool">[{stamp}] <i>{html.escape(note)}</i></div>')
                elif s.startswith('This session is being continued from a previous conversation'):
```

Bash Rebuild PDF and screenshot first page

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)" && "/c/Program Files/Python313/python.exe"
conversation_to_pdf.py "C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/4cfc3c23-d9ff-4995-bdcb-
9adeea99fe25.jsonl" "../../Claude Conversations/Tara 11-11-26 Conversation 1 (Tape spring L001 0-0-0 - axial, bending and
coiling Hashin checks, mass, PDF report).pdf" "Tara 11-11-26 Conversation 1: Tape spring L001 [0/0/0] - axial, bending and
coiling Hashin checks, mass, PDF report" 2>&1 | tail -1; grep -c "task-notification" _conv_pdf/conversation.html;
"/c/Program Files/Google/Chrome/Application/chrome.exe" --headless=new --disable-gpu --user-data-
dir="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26 (Tara)\\_conv_pdf\\chrome_profile" --window-
size=850,1500 --screenshot="F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26
(Tara)\\_conv_pdf\\shot.png" "file:///F:/Tara/BenazzoVenkatadri_ThinShellSims/Code%20for%20Project/11-11-
26%20(Tara)/_conv_pdf/conversation.html" 2>&1 | tail -1
```

```
The file F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\conversation_to_pdf.py has been updated
successfully.
```